

## CAMBRIDGE INTERNATIONAL AS &amp; A LEVEL CHEMISTRY (9701)

**Topic 3 — Chemical Bonding (Paper 1 Multiple Choice)**

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions &amp; Detailed Solutions

|                |                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------|
| CANDIDATE      | Urwah                                                                                                                    |
| SYLLABUS       | Cambridge International AS Level Chemistry (9701)                                                                        |
| COMPONENT      | Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)                                                               |
| TOPIC COVERAGE | 3.1 Electronegativity · 3.2 Ionic Lattices · 3.3 Metallic Bonding · 3.4 Covalent & Dative · 3.5 Shapes & VSEPR · 3.6 Int |
| TOTAL VOLUME   | 110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)                                                        |

**HOW TO USE THIS PAPER 1 MCQ PACK**

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

**SYLLABUS SUBTOPICS COVERED IN THIS PACK****3.1 Electronegativity & Bonding**

— Pauling scale, periodic trends across periods and down groups, bond dipole moments, symmetrical cancellation

**3.2 Ionic Bonding & Giant Lattices**

— Non-directional electrostatic attractions, lattice energy factors (charge and radius), NaCl 6:6 and CsCl 8:8 lattices

**3.3 Metallic Bonding & Properties**

— Electrostatic attraction to delocalised sea of electrons, Period 3 strength trends (Na &lt; Mg &lt; Al), thermal and elect

**3.4 Covalent & Coordinate (Dative) Bonding**— Shared electron pairs, orbital overlap (sigma vs pi), bond enthalpy vs length, coordinate bonds in NH<sub>4</sub><sup>+</sup>, H<sub>3</sub>O<sup>+</sup>, Al**3.5 Shapes of Molecules & VSEPR Theory**

— Electron pair repulsion hierarchy, bond angle compression (~2.5° per lone pair), linear (180°), trigonal planar (120°)

**3.6 Intermolecular Forces & Hydrogen Bonding**

— London dispersion forces (instantaneous-induced dipoles, polarisability, surface area), permanent dipole-dipole

**3.7 Dot-and-Cross Diagrams & Lewis Structures**— Octet rule, electron-deficient molecules (BF<sub>3</sub>), expanded octet hypervalent species (PCl<sub>5</sub>, SF<sub>6</sub>), polyatomic ions,**High-Frequency Core Repeats**

— The 10 most frequently examined Cambridge Paper 1 questions on Chemical Bonding (Q101 – Q110).

## SUBTOPIC 3.1 — ELECTRONEGATIVITY &amp; BOND POLARITY (Q1 – Q14)

**1** [EASY] Definition of Electronegativity — 9701/11/M/J/23/Q9

[1]

Which statement correctly defines electronegativity according to the Pauling scale?

- [A] The energy released when an isolated gaseous atom acquires an electron to form a uni-negative ion.
- [B] The power of an atom in a molecule to attract the shared pair of electrons in a covalent bond towards itself.
- [C] The electrostatic attraction between a positively charged atomic nucleus and surrounding delocalised electrons.
- [D] The energy required to remove one mole of electrons from one mole of gaseous atoms under standard conditions.

**2** [EASY] Electronegativity Trend Across Period 3 — 9701/12/M/J/23/Q10

[1]

How and why does Pauling electronegativity change from sodium to chlorine across Period 3?

- [A] It decreases because the atomic radius decreases, increasing electron-electron repulsion in the valence shell.
- [B] It increases because the number of occupied principal quantum shells increases across the period.
- [C] It decreases because the metallic character changes to non-metallic character across the period.
- [D] It increases because nuclear charge increases while shielding remains approximately constant, attracting bonding electrons more strongly.

**3** [EASY] Electronegativity Trend Down Group 17 — 9701/13/O/N/23/Q9

[1]

Which sequence correctly ranks the halogens in order of decreasing electronegativity (highest to lowest)?

- [A] Fluorine > Chlorine > Bromine > Iodine
- [B] Iodine > Bromine > Chlorine > Fluorine
- [C] Chlorine > Fluorine > Bromine > Iodine
- [D] Fluorine > Bromine > Chlorine > Iodine

**4** [HARD] Polar Bonds in Symmetrical Non-Polar Molecules — 9701/11/O/N/23/Q11

[1]

Which molecule contains polar covalent bonds but possesses zero overall molecular dipole moment?

- [A] Trichloromethane,  $\text{CHCl}_3$
- [B] Dichloromethane,  $\text{CH}_2\text{Cl}_2$
- [C] Tetrachloromethane,  $\text{CCl}_4$
- [D] Chloromethane,  $\text{CH}_3\text{Cl}$

**5** [HARD] Molecular Dipoles in Triatomic Molecules — 9701/12/F/M/24/Q8

[1]

Both carbon dioxide,  $\text{CO}_2$ , and sulfur dioxide,  $\text{SO}_2$ , contain polar bonds between non-metal atoms and oxygen.Why is  $\text{CO}_2$  non-polar whereas  $\text{SO}_2$  is a polar molecule?

- [A]  $\text{CO}_2$  is linear so its two equal and opposite bond dipoles cancel, whereas  $\text{SO}_2$  is non-linear (bent) so its bond dipoles reinforce.
- [B] Carbon is more electronegative than sulfur, resulting in weaker bond polarities in  $\text{CO}_2$ .
- [C]  $\text{SO}_2$  contains coordinate dative bonds that induce an asymmetric electron distribution.
- [D]  $\text{CO}_2$  has a giant covalent lattice whereas  $\text{SO}_2$  consists of discrete simple molecules.

**6** [HARD] Continuum of Chemical Bonding — 9701/12/M/J/22/Q10

[1]

Which pair of bonded atoms has the greatest degree of ionic character based on Pauling electronegativity values?

[Pauling values: Cs = 0.79, Na = 0.93, H = 2.20, Cl = 3.16, F = 3.98]

- [A] Na – Cl
- [B] H – F
- [C] Na – F
- [D] Cs – F

**7 [HARD]** Covalent Character in Ionic Compounds (Fajan's Rules) — 9701/13/M/J/23/Q11**[1]**

Which metal halide exhibits the greatest degree of covalent character due to polarisation of the anion?

- [A] Sodium fluoride, NaF
- [B] Aluminum iodide, AlI<sub>3</sub>
- [C] Magnesium fluoride, MgF<sub>2</sub>
- [D] Aluminum fluoride, AlF<sub>3</sub>

**8 [EASY]** Bond Dipoles in Boron Trifluoride — 9701/11/F/M/23/Q9**[1]**

Boron trifluoride, BF<sub>3</sub>, contains three strongly polar B-F bonds (electronegativity: B = 2.04, F = 3.98). What is the net molecular dipole moment of BF<sub>3</sub>?

- [A] Non-zero, because fluorine is significantly more electronegative than boron.
- [B] Zero, because the B-F bonds are non-polar due to equal sharing of electrons.
- [C] Zero, because the trigonal planar geometry arranges the three equal bond dipoles at 120°, cancelling each other out completely.
- [D] Non-zero, because boron has a vacant 2p orbital that distorts the molecular symmetry.

**9 [EASY]** Polarity of Haloalkanes — 9701/12/O/N/22/Q12**[1]**

Which chloromethane molecule has the greatest net molecular dipole moment?

- [A] Dichloromethane, CH<sub>2</sub>Cl<sub>2</sub>
- [B] Chloromethane, CH<sub>3</sub>Cl
- [C] Trichloromethane, CHCl<sub>3</sub>
- [D] Tetrachloromethane, CCl<sub>4</sub>

**10 [HARD]** Polarising Power of Alkaline Earth Cations — 9701/11/M/J/24/Q10**[1]**

Why is anhydrous beryllium chloride, BeCl<sub>2</sub>, predominantly covalent while barium chloride, BaCl<sub>2</sub>, is a typical giant ionic lattice?

- [A] Be<sup>2+</sup> has an exceptionally small ionic radius and high charge density, strongly polarising the chloride electron clouds to share electrons.
- [B] Beryllium has a lower first ionisation energy than barium, making electron transfer unfavorable.
- [C] Chloride ions are more polarisable in the presence of large Ba<sup>2+</sup> cations.
- [D] Barium forms covalent coordinate bonds with chloride ligands due to empty 5d orbitals.

**11 [HARD]** Electronegativity and Bond Enthalpy in Hydrogen Halides — 9701/13/O/N/22/Q11**[1]**

The Pauling electronegativities of halogens decrease down Group 17: F (3.98), Cl (3.16), Br (2.85), I (2.66).

Which statement correctly links electronegativity to the thermal stability of hydrogen halides (HF, HCl, HBr, HI)?

- [A] HI has the highest thermal stability because iodine has the largest electron cloud, maximizing London dispersion forces.
- [B] Thermal stability is independent of electronegativity and depends solely on molecular mass.
- [C] HBr is more thermally stable than HCl because bromine has a lower electronegativity, reducing bond strain.
- [D] HF has the highest thermal stability because high electronegativity difference results in a short, strongly polar bond with high bond dissociation energy.

**12 [EASY]** Electronegativity of Nitrogen vs Phosphorus — 9701/12/M/J/24/Q9**[1]**

Nitrogen (electronegativity = 3.04) and phosphorus (electronegativity = 2.19) are both Group 15 elements.

Which statement regarding their hydrides, NH<sub>3</sub> and PH<sub>3</sub>, is correct?

- [A] PH<sub>3</sub> has a much larger molecular dipole moment than NH<sub>3</sub> because phosphorus is larger.
- [B] Both NH<sub>3</sub> and PH<sub>3</sub> form extensive intermolecular hydrogen bonds in the liquid state.
- [C] The N-H bonds in NH<sub>3</sub> are significantly more polar than the P-H bonds in PH<sub>3</sub>.
- [D] The bond angle in NH<sub>3</sub> is smaller than that in PH<sub>3</sub> because nitrogen attracts bonding pairs closer.

**13 [HARD]** Dipole Moment of cis- and trans-1,2-Dichloroethene — 9701/11/O/N/24/Q12**[1]**

Consider the geometric isomers of 1,2-dichloroethene: cis-1,2-dichloroethene and trans-1,2-dichloroethene.

Which statement concerning their molecular polarities is correct?

- [A] The cis isomer has a permanent dipole moment, while the trans isomer has zero net dipole moment because its C-Cl bond dipoles cancel.
- [B] Both isomers are completely non-polar because the C=C double bond is non-polar.
- [C] The trans isomer has a larger dipole moment because the chlorine atoms are at maximum distance from each other.
- [D] The cis isomer has zero dipole moment because the two chlorine atoms are on the same side of the double bond.

**14 [EASY]** Electronegativity and Oxidation State Assignment — 9701/13/M/J/24/Q11**[1]**

In oxygen difluoride, OF<sub>2</sub>, what is the oxidation state of oxygen, and which concept explains this value?

- [A] -2, because oxygen is in Group 16 and always adopts an oxidation state of -2 in compounds.
- [B] +1, because each fluorine atom carries a +1 formal charge.
- [C] +2, because fluorine is more electronegative than oxygen and attracts shared electrons more strongly.
- [D] -1, because OF<sub>2</sub> is a peroxide with an O-O single bond.

**SUBTOPIC 3.2 — IONIC BONDING & GIANT LATTICES (Q15 – Q28)****15 [EASY]** Fundamental Nature of the Ionic Bond — 9701/11/M/J/23/Q11**[1]**

Which statement describes the fundamental nature of an ionic bond?

- [A] The sharing of electron pairs localized between two positively charged nuclei.
- [B] The non-directional electrostatic attraction between oppositely charged ions extending throughout a giant three-dimensional lattice.
- [C] The electrostatic attraction between positive metal cations and a sea of delocalised electrons.
- [D] The directional electrostatic attraction between permanent molecular dipoles.

**16 [EASY]** Sodium Chloride Lattice Coordination — 9701/12/M/J/23/Q12**[1]**

In the giant ionic lattice of crystalline sodium chloride, NaCl, what is the coordination number of each Na<sup>+</sup> ion and each Cl<sup>-</sup> ion?

- [A] Each Na<sup>+</sup> is coordinated to 8 Cl<sup>-</sup> ions, and each Cl<sup>-</sup> is coordinated to 8 Na<sup>+</sup> ions (8:8 coordination).
- [B] Each Na<sup>+</sup> is coordinated to 4 Cl<sup>-</sup> ions, and each Cl<sup>-</sup> is coordinated to 4 Na<sup>+</sup> ions (4:4 coordination).
- [C] Each Na<sup>+</sup> is coordinated to 12 Cl<sup>-</sup> ions, and each Cl<sup>-</sup> is coordinated to 6 Na<sup>+</sup> ions (12:6 coordination).
- [D] Each Na<sup>+</sup> is coordinated to 6 Cl<sup>-</sup> ions, and each Cl<sup>-</sup> is coordinated to 6 Na<sup>+</sup> ions (6:6 coordination).

**17 [HARD]** Factors Governing Lattice Enthalpy Magnitude — 9701/11/O/N/23/Q13**[1]**

Lattice enthalpy is a quantitative measure of ionic bond strength in a crystal lattice. Which combination of ionic properties produces the most exothermic lattice enthalpy?

- [A] Low ionic charges and large ionic radii.
- [B] High ionic charges and large ionic radii.
- [C] High ionic charges and small ionic radii.
- [D] Low ionic charges and small ionic radii.

**18 [HARD]** Comparing Melting Points of Giant Ionic Solids — 9701/13/O/N/23/Q12**[1]**

The melting point of magnesium oxide, MgO, is 2852 °C, whereas that of sodium chloride, NaCl, is 801 °C.

Which statement best explains this immense difference in melting points?

- [A] Mg<sup>2+</sup> and O<sup>2-</sup> ions carry double the charge of Na<sup>+</sup> and Cl<sup>-</sup> ions and have smaller ionic radii, resulting in vastly stronger electrostatic attraction.
- [B] MgO forms a giant covalent macromolecular network while NaCl forms a giant ionic lattice.
- [C] Magnesium has a higher electronegativity than sodium, causing MgO to be predominantly covalent.
- [D] NaCl decomposes before melting due to weak intermolecular attractions between ion pairs.

**19** [EASY] Electrical Conductivity of Ionic Compounds — 9701/12/F/M/24/Q10

[1]

Under which conditions does an ionic compound such as potassium bromide, KBr, conduct an electric current?

- [A] In the solid state, molten state, and aqueous solution.
- [B] In the solid state and molten state, but not in aqueous solution.
- [C] Only when dissolved in non-polar organic solvents.
- [D] When molten or dissolved in aqueous solution, but not in the solid state.

**20** [HARD] Brittleness and Cleavage of Ionic Crystals — 9701/13/M/J/22/Q12

[1]

When an ionic crystal such as calcium fluoride,  $\text{CaF}_2$ , is struck sharply with a hammer, it shatters along smooth planes rather than deforming.

What is the scientific explanation for this brittleness?

- [A] Delocalised electrons are forced into localized atomic orbitals, destroying metallic cohesion.
- [B] The impact forces layers of ions to shift, bringing ions of like charge into direct alignment, causing intense electrostatic repulsion that cleaves the crystal.
- [C] Strong covalent bonds along the crystal axes undergo homolytic cleavage simultaneously.
- [D] The crystal lattice contains micro-voids that trap air and trigger explosive decompression.

**21** [HARD] Lattice Enthalpy Trend in Group 2 Oxides — 9701/11/M/J/24/Q11

[1]

Which sequence correctly displays the Group 2 oxides in order of decreasing melting point (highest to lowest)?

[Atomic numbers: Mg = 12, Ca = 20, Sr = 38, Ba = 56]

- [A]  $\text{BaO} > \text{SrO} > \text{CaO} > \text{MgO}$
- [B]  $\text{MgO} > \text{CaO} > \text{SrO} > \text{BaO}$
- [C]  $\text{CaO} > \text{MgO} > \text{BaO} > \text{SrO}$
- [D]  $\text{MgO} > \text{BaO} > \text{CaO} > \text{SrO}$

**22** [HARD] Solubility of Ionic Solids in Water — 9701/12/O/N/23/Q14

[1]

For an ionic solid to dissolve spontaneously in water, which thermodynamic condition is generally required?

- [A] The lattice enthalpy must be vastly more exothermic than the sum of the hydration enthalpies.
- [B] The solvent must form strong covalent bonds with the cations.
- [C] The ions must undergo oxidation and reduction reactions with water molecules.
- [D] The hydration enthalpies of the ions must be sufficiently exothermic to compensate for the endothermic lattice breaking process.

**23** [EASY] Lattice Energy of Alkali Metal Halides — 9701/13/M/J/24/Q12

[1]

Which sodium halide has the most exothermic lattice enthalpy?

- [A] Sodium fluoride, NaF
- [B] Sodium chloride, NaCl
- [C] Sodium bromide, NaBr
- [D] Sodium iodide, NaI

**24** [EASY] Ionic vs Covalent Properties Comparison — 9701/11/F/M/22/Q11

[1]

Substance X is a crystalline white solid at room temperature with a melting point of  $772^\circ\text{C}$ . It does not conduct electricity as a solid, but dissolves readily in water to give a solution that conducts electricity efficiently. What is the identity of X?

- [A] Silicon dioxide,  $\text{SiO}_2$
- [B] Sucrose,  $\text{C}_{12}\text{H}_{22}\text{O}_{11}$
- [C] Calcium chloride,  $\text{CaCl}_2$
- [D] Graphite, C

**25 [HARD]** Caesium Chloride Lattice Structure — 9701/12/M/J/22/Q13**[1]**

Caesium chloride, CsCl, adopts a body-centred cubic-like lattice structure rather than the sodium chloride, NaCl, rock-salt structure. What is the coordination number of Cs<sup>+</sup> and Cl<sup>-</sup> in CsCl?

- [A] 8:8 coordination
- [B] 6:6 coordination
- [C] 4:4 coordination
- [D] 12:12 coordination

**26 [HARD]** Electron Density Map Evidence for Discrete Ions — 9701/13/O/N/22/Q13**[1]**

In an X-ray diffraction electron density contour map of crystalline sodium chloride, what feature provides direct experimental evidence that NaCl consists of discrete ions rather than shared covalent electron pairs?

- [A] The electron density is uniformly distributed across the entire crystal volume.
- [B] There are bridges of continuous high electron density connecting adjacent nuclei.
- [C] The electron density around the chlorine nucleus is identical to that around the sodium nucleus.
- [D] The electron density between adjacent sodium and chlorine nuclei drops to zero.

**27 [EASY]** Ionic Formula Deduction from Ion Charges — 9701/11/O/N/24/Q14**[1]**

An element M in Group 2 reacts with a non-metal X in Group 15. What is the empirical formula of the resulting ionic compound?

- [A] MX
- [B] M3X2
- [C] M2X3
- [D] M3X

**28 [HARD]** Charge on Transition Metal Complex Ion — 9701/12/M/J/24/Q13**[1]**

In the ionic compound potassium hexacyanoferrate(II), K<sub>4</sub>[Fe(CN)<sub>6</sub>], what is the charge on the complex anion and the oxidation state of iron?

- [A] Charge on anion = 4+; oxidation state of Fe = +2
- [B] Charge on anion = 3-; oxidation state of Fe = +3
- [C] Charge on anion = 4-; oxidation state of Fe = +2
- [D] Charge on anion = 2-; oxidation state of Fe = +4

**SUBTOPIC 3.3 — METALLIC BONDING & PHYSICAL PROPERTIES (Q29 – Q42)****29 [EASY]** Definition of Metallic Bonding — 9701/11/M/J/23/Q12**[1]**

Which phrase accurately defines the nature of metallic bonding?

- [A] The electrostatic attraction between oppositely charged discrete cations and anions.
- [B] The strong electrostatic attraction between a regular lattice of positive metal ions and a 'sea' of delocalised valence electrons.
- [C] The sharing of localized pairs of electrons between adjacent metal nuclei.
- [D] The dipole-dipole attractions between polar metallic molecules.

**30 [HARD]** Trend in Metallic Bond Strength Across Period 3 — 9701/12/M/J/23/Q13**[1]**

The melting points of the Period 3 metals increase sharply from sodium to magnesium and aluminum:  
Na (98 °C) < Mg (650 °C) < Al (660 °C)

Which factor accounts for this increase in metallic bond strength?

- [A] The charge on the metal cation increases (+1 to +2 to +3), cation radius decreases, and more delocalised electrons are released per atom.
- [B] The number of occupied electron shells increases from sodium to aluminum.
- [C] The covalent bond character increases due to greater electronegativity differences.
- [D] The crystal structure changes from giant metallic to giant covalent macromolecular.

**31** [EASY] Electrical Conductivity Mechanism in Metals — 9701/13/M/J/23/Q13

[1]

Why are metals excellent conductors of electricity in both solid and molten states?

- [A] The positive metal cations migrate rapidly towards the negative electrode.
- [B] Thermal vibrations break covalent bonds and generate free electron-hole pairs.
- [C] Electrons are transferred between localized ionic bonds along crystal boundaries.
- [D] The delocalised valence electrons are free to drift through the lattice when a potential difference is applied.

**32** [HARD] Effect of Temperature on Metallic Conductivity — 9701/11/O/N/23/Q14

[1]

As the temperature of a copper wire increases, what happens to its electrical conductivity, and why?

- [A] Conductivity increases because higher thermal energy releases more valence electrons into the delocalised pool.
- [B] Conductivity remains unchanged because the number of delocalised electrons is strictly constant.
- [C] Conductivity decreases because increased thermal vibration of the metal cations scatters the drifting delocalised electrons.
- [D] Conductivity increases because the cations expand and reduce electrical resistance.

**33** [EASY] Malleability and Ductility of Metals — 9701/12/O/N/23/Q15

[1]

Pure metals such as gold and copper are malleable (can be hammered into sheets) and ductile (can be drawn into wires). What structural feature enables this deformation without fracturing?

- [A] Layers of metal cations can slide over each other without disrupting the non-directional metallic bonding maintained by the delocalised electrons.
- [B] Individual covalent bonds stretch and re-form instantaneously between adjacent atoms.
- [C] The metal cations carry zero net charge so no electrostatic repulsion is generated upon sliding.
- [D] The presence of microscopic voids allows atoms to collapse into unoccupied lattice sites.

**34** [HARD] Hardness of Alloys vs Pure Metals — 9701/13/O/N/23/Q15

[1]

Brass is an alloy consisting of copper (approx 70%) and zinc (approx 30%). Why is brass significantly harder and less malleable than pure copper?

- [A] Zinc forms rigid localized covalent bonds with copper that lock the crystal structure.
- [B] Zinc atoms react with copper to form a brittle giant ionic lattice.
- [C] Zinc atoms have a different atomic radius from copper atoms, disrupting the regular lattice layers and preventing them from sliding easily.
- [D] Zinc removes delocalised electrons from the metallic pool, reducing metallic bond ductility.

**35** [EASY] Melting Point Trend Down Group 1 Metals — 9701/11/F/M/24/Q11

[1]

The melting points of alkali metals decrease steadily down Group 1:

Li (181 °C) > Na (98 °C) > K (63 °C) > Rb (39 °C) > Cs (29 °C)

What is the explanation for this downward trend?

- [A] The number of delocalised electrons per atom decreases down the group.
- [B] Cation radius increases down the group, increasing the distance between the nucleus and delocalised electrons, weakening the metallic bond.
- [C] The electronegativity increases down the group, promoting covalent bonding.
- [D] The screening effect decreases down the group, allowing outer electrons to escape.

**36** [EASY] Thermal Conductivity of Metals — 9701/12/F/M/23/Q12

[1]

Which two mechanisms contribute to the exceptionally high thermal conductivity of metals?

- [A] Covalent bond cleavage and immediate reformation across thermal gradients.
- [B] Electrochemical oxidation and reduction cycles occurring across the metal bulk.
- [C] Convection currents created by the physical movement of metal cations through the crystal.
- [D] Rapid kinetic energy transfer by mobile delocalised electrons and lattice vibrational waves (phonons).

**37 [HARD]** Comparison of Conductivity Across Different Solids — 9701/13/F/M/23/Q10**[1]**

Consider three solid elements: aluminium, silicon, and sulfur. Which sequence correctly lists them in order of decreasing electrical conductivity at room temperature?

- [A] Silicon > Aluminium > Sulfur
- [B] Sulfur > Silicon > Aluminium
- [C] Aluminium > Silicon > Sulfur
- [D] Aluminium > Sulfur > Silicon

**38 [HARD]** Metallic Bonding in Transition Metals — 9701/11/M/J/22/Q14**[1]**

Transition metals such as iron and vanadium have significantly higher melting points and tensile strengths than Group 1 and Group 2 metals. Why is metallic bonding so much stronger in transition metals?

- [A] Both 3d and 4s electrons participate in the delocalised metallic bond, providing a much higher number of delocalised electrons per atom.
- [B] Transition metals form localized covalent coordinate bonds between adjacent metal atoms.
- [C] The atomic nuclei of transition metals carry lower positive charges, reducing electron repulsion.
- [D] Transition metals adopt simple hexagonal open lattices that maximize covalent resonance.

**39 [HARD]** Liquid State of Mercury at Room Temperature — 9701/12/M/J/24/Q14**[1]**

Mercury (Hg,  $Z = 80$ ) is the only metal that is liquid at room temperature (melting point  $-39\text{ }^{\circ}\text{C}$ ). Which factor contributes to the unusually weak metallic bonding in mercury?

- [A] Mercury contains no valence electrons in its outermost shell.
- [B] Mercury atoms are held together exclusively by weak London dispersion forces.
- [C] Mercury has an exceptionally small atomic radius that prevents orbital overlap.
- [D] The 6s2 valence electrons are tightly held in a stable, relativistic contracted shell and participate poorly in metallic delocalisation.

**40 [HARD]** Close-Packed Structures in Metals — 9701/13/O/N/24/Q15**[1]**

Most metallic elements have high densities because their cations pack tightly in close-packed crystal structures. What is the coordination number of an atom in a face-centred cubic (fcc) or hexagonal close-packed (hcp) metal lattice?

- [A] 8
- [B] 12
- [C] 6
- [D] 4

**41 [HARD]** Electrical Conduction in Molten Sodium Chloride vs Liquid Aluminium — 9701/11/O/N/24/Q16**[1]**

Both molten sodium chloride and liquid aluminium conduct electricity. What are the mobile charge carriers responsible for conduction in each liquid?

- [A] Molten NaCl: mobile electrons; Liquid Al: mobile  $\text{Al}^{3+}$  ions.
- [B] Molten NaCl: mobile  $\text{Na}^{+}$  and  $\text{Cl}^{-}$  ions; Liquid Al: mobile delocalised electrons.
- [C] Molten NaCl: mobile  $\text{Na}^{+}$  ions only; Liquid Al: mobile  $\text{Al}^{3+}$  ions only.
- [D] Both conduct exclusively via the transport of delocalised electrons.

**42 [EASY]** Photoelectric Effect and Metallic Bonding — 9701/12/M/J/22/Q15**[1]**

When clean zinc metal is irradiated with ultraviolet light, electrons are emitted from its surface. What does this photoelectric phenomenon demonstrate about the electrons in a metal?

- [A] Zinc atoms undergo spontaneous alpha decay upon exposure to ultraviolet light.
- [B] Covalent bonds in zinc are cleaved homolytically by low-frequency infrared photons.
- [C] Electrons in the inner 1s shell are excited into the conduction band.
- [D] Valence electrons in metals are delocalised and can be ejected when photon energy exceeds the metal's work function.

## SUBTOPIC 3.4 — COVALENT &amp; COORDINATE (DATIVE) BONDING (Q43 – Q64)

43 [EASY] Definition of a Covalent Bond — 9701/11/M/J/23/Q14

[1]

What constitutes a single covalent bond according to modern electronic theory?

- [A] The electrostatic attraction between the nuclei of two bonded atoms and a shared pair of electrons localized between them.
- [B] The complete transfer of one electron from a metal atom to a non-metal atom.
- [C] The attraction between permanent molecular dipoles of adjacent molecules.
- [D] The interaction between an unhybridized p orbital and a filled d orbital.

44 [EASY] Orbital Overlap: Sigma (sigma) Bonds — 9701/12/M/J/23/Q14

[1]

How is a sigma (sigma) covalent bond formed between two atoms?

- [A] By the sideways (lateral) overlap of parallel unhybridized p orbitals above and below the internuclear axis.
- [B] By the donation of a lone pair into an empty d orbital of a transition metal.
- [C] By the direct 'head-on' (axial) overlap of atomic orbitals along the internuclear axis.
- [D] By the electrostatic attraction between delocalised electrons and positive cations.

45 [EASY] Orbital Overlap: Pi (pi) Bonds — 9701/13/M/J/23/Q14

[1]

How is a pi (pi) covalent bond formed between two carbon atoms in an alkene?

- [A] By the sideways (lateral) overlap of two parallel, unhybridized p orbitals positioned perpendicular to the internuclear axis.
- [B] By the axial head-on overlap of two 2s atomic orbitals.
- [C] By the promotion of an electron from a 2s orbital into an excited 3s orbital.
- [D] By the donation of a non-bonding lone pair from one carbon atom to another.

46 [HARD] Bond Strength Comparison: Sigma vs Pi Bonds — 9701/11/O/N/23/Q15

[1]

The C-C single bond energy in ethane is  $348 \text{ kJ mol}^{-1}$ , while the C=C double bond energy in ethene is  $612 \text{ kJ mol}^{-1}$ .

Why is the second bond (the pi bond) weaker than the first bond (the sigma bond)?

- [A] Pi electrons are located closer to the carbon nuclei, experiencing greater repulsion.
- [B] The pi bond is formed by the overlap of higher energy d orbitals.
- [C] The sigma bond contains two electron pairs while the pi bond contains only one electron pair.
- [D] Sideways overlap of parallel p orbitals is less extensive than the direct head-on axial overlap of orbitals in a sigma bond.

47 [EASY] Counting Sigma and Pi Bonds in Ethyne — 9701/12/O/N/23/Q16

[1]

How many sigma (sigma) bonds and pi (pi) bonds are present in a single molecule of ethyne,  $\text{HC}\equiv\text{CH}$ ?

- [A] 2 sigma bonds and 3 pi bonds
- [B] 3 sigma bonds and 2 pi bonds
- [C] 1 sigma bond and 2 pi bonds
- [D] 5 sigma bonds and 0 pi bonds

48 [HARD] Counting Sigma and Pi Bonds in Propenenitrile — 9701/13/O/N/23/Q16

[1]

Propenenitrile (acrylonitrile) has the structural formula  $\text{CH}_2=\text{CH}-\text{C}\equiv\text{N}$ .

What is the total number of sigma (sigma) bonds and pi (pi) bonds in this molecule?

- [A] 5 sigma bonds and 4 pi bonds
- [B] 7 sigma bonds and 2 pi bonds
- [C] 6 sigma bonds and 3 pi bonds
- [D] 6 sigma bonds and 2 pi bonds

**49 [HARD]** Bond Length and Bond Strength in Nitrogen Compounds — 9701/11/F/M/24/Q12**[1]**

Consider the three nitrogen-nitrogen bonds:

1: Single N-N bond in hydrazine, N<sub>2</sub>H<sub>4</sub>

2: Double N=N bond in diazene, N<sub>2</sub>H<sub>2</sub>

3: Triple N≡N bond in nitrogen gas, N<sub>2</sub>

Which statement correctly compares their bond lengths and bond enthalpies?

- [A]** N<sub>2</sub>H<sub>4</sub> has the shortest bond length and greatest bond enthalpy because hydrogen atoms stabilize the bond.
- [B]** N<sub>2</sub> has the shortest bond length and greatest bond enthalpy; N<sub>2</sub>H<sub>4</sub> has the longest bond length and smallest bond enthalpy.
- [C]** Bond length increases from single to triple bond as more electrons are shared.
- [D]** All three bonds have identical lengths because the nitrogen atoms have the same covalent radius.

**50 [EASY]** Definition and Formation of a Coordinate (Dative) Bond — 9701/12/F/M/24/Q13**[1]**

What characterizes a coordinate (dative covalent) bond?

- [A]** A covalent bond in which both electrons of the shared pair are supplied by the same atom.
- [B]** An electrostatic attraction between a metal cation and an anion.
- [C]** A covalent bond formed by the overlap of two singly occupied atomic orbitals.
- [D]** A weak intermolecular force between a hydrogen atom and a fluorine atom.

**51 [EASY]** Coordinate Bonding in the Ammonium Ion, NH<sub>4</sub><sup>+</sup> — 9701/13/F/M/24/Q12**[1]**

When ammonia, NH<sub>3</sub>, reacts with an acid, it forms an ammonium ion, NH<sub>4</sub><sup>+</sup>.

Which statement concerning the bonding in NH<sub>4</sub><sup>+</sup> is correct?

- [A]** The coordinate bond is significantly longer and weaker than the three original covalent bonds.
- [B]** The ammonium ion contains two covalent bonds and two coordinate bonds.
- [C]** The nitrogen atom donates an electron to H<sup>+</sup>, forming an ionic bond between NH<sub>3</sub><sup>+</sup> and H<sup>-</sup>.
- [D]** One N-H bond is formed by a dative bond from the nitrogen lone pair to H<sup>+</sup>, but once formed, all four N-H bonds are identical in length and strength.

**52 [EASY]** Coordinate Bonding in the Hydronium Ion, H<sub>3</sub>O<sup>+</sup> — 9701/11/M/J/22/Q16**[1]**

How is the coordinate bond formed in the hydroxonium (hydronium) ion, H<sub>3</sub>O<sup>+</sup>?

- [A]** An electron is transferred from oxygen to H<sup>+</sup>, forming an ionic pair.
- [B]** A hydrogen atom donates its electron into a vacant 2p orbital of the oxygen atom.
- [C]** One of the two lone pairs on the oxygen atom of a water molecule is donated into the vacant 1s orbital of a hydrogen ion, H<sup>+</sup>.
- [D]** Two water molecules share a proton via a hydrogen bond.

**53 [HARD]** Coordinate Bonding in Aluminium Chloride Dimer, Al<sub>2</sub>Cl<sub>6</sub> — 9701/12/M/J/22/Q16**[1]**

At temperatures below 200 °C, aluminium chloride exists as a gaseous dimer, Al<sub>2</sub>Cl<sub>6</sub>.

How is this dimer held together?

- [A]** Two chlorine atoms each donate a lone pair into an empty 3p orbital of an adjacent aluminium atom, forming two dative covalent bridging bonds.
- [B]** The dimer is held together by weak London dispersion forces between two planar AlCl<sub>3</sub> molecules.
- [C]** An aluminium atom donates two electrons to another aluminium atom, forming an Al-Al metallic bond.
- [D]** Six chlorine atoms form a giant ionic lattice around two central aluminium cations.

**54 [HARD]** Coordinate Bonding in Carbon Monoxide, CO — 9701/13/M/J/22/Q15**[1]**

Carbon monoxide, CO, has a triple bond between carbon and oxygen. What types of covalent bonds make up this triple bond?

- [A] Three ordinary covalent bonds formed by equal sharing of three electron pairs.
- [B] One covalent bond and two coordinate bonds where carbon donates electrons to oxygen.
- [C] Two ordinary covalent bonds (one sigma, one pi) and one coordinate (dative) bond where oxygen donates a lone pair to carbon.
- [D] One ionic bond and two covalent pi bonds.

**55 [HARD]** Boron Trifluoride-Ammonia Adduct, F<sub>3</sub>B-NH<sub>3</sub> — 9701/11/O/N/22/Q15**[1]**

When gaseous boron trifluoride, BF<sub>3</sub>, reacts with gaseous ammonia, NH<sub>3</sub>, a solid adduct F<sub>3</sub>B-NH<sub>3</sub> is formed.

Which statement correctly describes the change in geometry around the boron atom during this reaction?

- [A] Boron changes from tetrahedral (109.5°) to trigonal planar (120°) as it loses an electron.
- [B] Boron changes from trigonal planar (120°) to tetrahedral (approx 109.5°) as it accepts a lone pair from nitrogen.
- [C] Boron remains trigonal planar because the coordinate bond does not alter electron pair repulsion.
- [D] Boron changes from linear (180°) to bent (104.5°) upon coordinate bond formation.

**56 [EASY]** Bond Enthalpy of Carbon-Halogen Bonds — 9701/12/O/N/22/Q16**[1]**

Which halogenoalkane has the most reactive (most easily cleaved) carbon-halogen bond during nucleophilic substitution?

- [A] 1-Bromobutane (C-Br bond enthalpy = 280 kJ mol<sup>-1</sup>)
- [B] 1-Chlorobutane (C-Cl bond enthalpy = 340 kJ mol<sup>-1</sup>)
- [C] 1-Fluorobutane (C-F bond enthalpy = 467 kJ mol<sup>-1</sup>)
- [D] 1-Iodobutane (C-I bond enthalpy = 240 kJ mol<sup>-1</sup>)

**57 [EASY]** Homolytic vs Heterolytic Bond Fission — 9701/13/O/N/22/Q16**[1]**

What occurs during the homolytic fission of a covalent bond, such as Cl-Cl in the presence of UV light?

- [A] The covalent bond breaks unsymmetrically, with one atom taking both electrons to form a cation and an anion.
- [B] A coordinate bond is formed between two chlorine molecules.
- [C] The covalent bond breaks symmetrically, with each bonded atom retaining one of the shared electrons to form two free radicals.
- [D] Electrons are transferred to the solvent, oxidizing the halogen.

**58 [HARD]** Restricted Rotation Around Double Bonds — 9701/11/F/M/23/Q13**[1]**

Why can molecules exhibit cis-trans (geometric) isomerism around a C=C double bond, but not around a C-C single bond?

- [A] Rotation around a C=C bond requires breaking the pi bond (loss of sideways p orbital overlap), which requires significant energy at room temperature.
- [B] The sigma bond in C=C is rigid and cannot twist, whereas pi bonds allow free rotation.
- [C] C=C double bonds contain coordinate bonds that lock the bonded carbon atoms in position.
- [D] Steric hindrance between carbon nuclei prevents thermal rotation entirely.

**59 [HARD]** Delocalisation of Pi Electrons in the Nitrate Ion, NO<sub>3</sub><sup>-</sup> — 9701/12/F/M/23/Q14**[1]**

Experimental measurements show that all three N-O bonds in the nitrate ion, NO<sub>3</sub><sup>-</sup>, have exactly identical bond lengths (124 pm), which are intermediate between a single N-O bond (140 pm) and a double N=O bond (120 pm).

What explains this observation?

- [A] The nitrate ion rapidly oscillates between three different resonant isomers.
- [B] The three bonds consist of two coordinate bonds and one ionic bond.
- [C] The nitrogen atom undergoes sp<sup>3</sup> hybridization, forming four equivalent sigma bonds.
- [D] The pi electron pair is delocalised equally over all three nitrogen-oxygen bonds, giving each bond a bond order of 1.33.

**60** [EASY] Coordinate Bonding in Transition Metal Complexes — 9701/13/F/M/23/Q13**[1]**

In the complex ion  $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ , what role does the water molecule play, and what type of bond connects it to the central copper ion?

- [A] Water acts as a Brønsted–Lowry acid transferring a proton to  $\text{Cu}^{2+}$ .
- [B] Water acts as a ligand (Lewis base) donating a lone pair to form a coordinate (dative) bond to  $\text{Cu}^{2+}$ .
- [C] Water forms an ionic bond through electrostatic attraction to  $\text{Cu}^{2+}$ .
- [D] Water forms a covalent sigma bond by sharing single unpaired electrons with  $\text{Cu}^{2+}$ .

**61** [EASY] Bond Enthalpy and Chemical Inertness of Nitrogen Gas — 9701/11/M/J/24/Q13**[1]**

Why is molecular nitrogen,  $\text{N}_2$ , remarkably unreactive at room temperature and pressure?

- [A] Nitrogen molecules are completely non-polar and cannot dissolve in any solvent.
- [B] The  $\text{N}\equiv\text{N}$  triple bond has an extraordinarily large bond enthalpy ( $945 \text{ kJ mol}^{-1}$ ), requiring huge activation energy to break.
- [C] Nitrogen has a high electronegativity that repels incoming electrophiles.
- [D] The nitrogen atom possesses no non-bonding lone pairs of electrons.

**62** [HARD] Bond Angle in Dative Adduct:  $\text{AlCl}_3$  vs  $\text{AlCl}_4^-$  — 9701/12/M/J/24/Q15**[1]**

What are the  $\text{Cl}-\text{Al}-\text{Cl}$  bond angles in aluminium chloride monomer,  $\text{AlCl}_3$ , and in the tetrachloroaluminate ion,  $\text{AlCl}_4^-$ ?

- [A]  $\text{AlCl}_3$ :  $109.5^\circ$ ;  $\text{AlCl}_4^-$ :  $120^\circ$
- [B]  $\text{AlCl}_3$ :  $107^\circ$ ;  $\text{AlCl}_4^-$ :  $104.5^\circ$
- [C]  $\text{AlCl}_3$ :  $180^\circ$ ;  $\text{AlCl}_4^-$ :  $90^\circ$
- [D]  $\text{AlCl}_3$ :  $120^\circ$ ;  $\text{AlCl}_4^-$ :  $109.5^\circ$

**63** [HARD] Bond Polarity vs Average Bond Enthalpy — 9701/13/M/J/24/Q14**[1]**

Which statement regarding covalent bond properties is correct?

- [A] A polar covalent bond is generally stronger than a purely covalent bond between similar atoms due to the additional electrostatic attraction between partial charges.
- [B] Bond enthalpy decreases as bond order increases.
- [C] Triple bonds are always longer than single bonds because they contain more electrons.
- [D] A homonuclear diatomic molecule such as  $\text{Cl}_2$  contains polar covalent bonds.

**64** [HARD] Hybridization in Ethene — 9701/11/O/N/24/Q17**[1]**

What is the hybridization state of the carbon atoms in ethene,  $\text{C}_2\text{H}_4$ , and what is the approximate  $\text{H}-\text{C}-\text{H}$  bond angle?

- [A]  $\text{sp}^3$  hybridized;  $109.5^\circ$
- [B]  $\text{sp}$  hybridized;  $180^\circ$
- [C]  $\text{sp}^2$  hybridized;  $120^\circ$
- [D]  $\text{sp}^3\text{d}$  hybridized;  $90^\circ$

**SUBTOPIC 3.5 — SHAPES OF MOLECULES & VSEPR THEORY (Q65 – Q80)****65** [EASY] Core Principle of VSEPR Theory — 9701/11/M/J/23/Q15**[1]**

Valence Shell Electron Pair Repulsion (VSEPR) theory predicts molecular geometry based on which fundamental principle?

- [A] Electron pairs around a central atom repel each other electrostatically and adopt positions of maximum separation to minimize repulsion.
- [B] Electrons attract each other to form stable pairs of opposite spin.
- [C] Atoms in a molecule vibrate along bond axes until they reach minimum kinetic energy.
- [D] Electronegative atoms always position themselves as close to the nucleus as possible.

**66** [EASY] Repulsion Hierarchy of Electron Pairs — 9701/12/M/J/23/Q15**[1]**

According to VSEPR theory, which sequence correctly ranks the strength of mutual repulsion between different electron pairs (greatest repulsion to least repulsion)?

- [A] Bonding pair – Bonding pair > Lone pair – Bonding pair > Lone pair – Lone pair
- [B] Lone pair – Bonding pair > Lone pair – Lone pair > Bonding pair – Bonding pair
- [C] Lone pair – Lone pair > Bonding pair – Bonding pair > Lone pair – Bonding pair
- [D] Lone pair – Lone pair > Lone pair – Bonding pair > Bonding pair – Bonding pair

**67** [HARD] Bond Angles in CH<sub>4</sub>, NH<sub>3</sub>, and H<sub>2</sub>O — 9701/13/M/J/23/Q15**[1]**

Methane (CH<sub>4</sub>), ammonia (NH<sub>3</sub>), and water (H<sub>2</sub>O) all have four electron pairs around their central atom.

Which sequence correctly lists their bond angles in decreasing order?

- [A] H<sub>2</sub>O (104.5°) > NH<sub>3</sub> (107°) > CH<sub>4</sub> (109.5°)
- [B] CH<sub>4</sub> (109.5°) > NH<sub>3</sub> (107°) > H<sub>2</sub>O (104.5°)
- [C] NH<sub>3</sub> (107°) > CH<sub>4</sub> (109.5°) > H<sub>2</sub>O (104.5°)
- [D] CH<sub>4</sub> (109.5°) > H<sub>2</sub>O (104.5°) > NH<sub>3</sub> (107°)

**68** [EASY] Shape and Bond Angle of Beryllium Chloride — 9701/11/O/N/23/Q17**[1]**

What is the shape and bond angle of a gaseous beryllium chloride, BeCl<sub>2</sub>, molecule?

- [A] Bent (non-linear), with a bond angle of 104.5°
- [B] Trigonal planar, with a bond angle of 120°
- [C] Linear, with a bond angle of 180°
- [D] Tetrahedral, with a bond angle of 109.5°

**69** [EASY] Shape and Bond Angle of Boron Trichloride — 9701/12/O/N/23/Q17**[1]**

What is the molecular geometry and bond angle of boron trichloride, BCl<sub>3</sub>?

- [A] Trigonal pyramidal, 107°
- [B] Trigonal planar, 120°
- [C] T-shaped, 90°
- [D] Tetrahedral, 109.5°

**70** [HARD] Shape of the Carbonate Ion, CO<sub>3</sub><sup>2-</sup> — 9701/13/O/N/23/Q17**[1]**

What is the shape and bond angle of the carbonate ion, CO<sub>3</sub><sup>2-</sup>?

- [A] Trigonal planar, 120°
- [B] Trigonal pyramidal, 107°
- [C] T-shaped, 90°
- [D] Tetrahedral, 109.5°

**71** [HARD] Shape and Bond Angle of Sulfur Dioxide, SO<sub>2</sub> — 9701/11/F/M/24/Q14**[1]**

What is the shape and approximate bond angle of a sulfur dioxide, SO<sub>2</sub>, molecule?

- [A] Linear, 180°
- [B] Trigonal planar, 120°
- [C] Tetrahedral, 109.5°
- [D] Bent (non-linear), approx 117° – 119°

**72 [HARD]** Shape and Bond Angles of Phosphorus Pentachloride,  $\text{PCl}_5$  — 9701/12/F/M/24/Q14**[1]**What is the shape and bond angles in a gaseous molecule of phosphorus pentachloride,  $\text{PCl}_5$ ?

- [A] Square pyramidal, with bond angles of  $90^\circ$
- [B] Pentagonal planar, with bond angles of  $72^\circ$
- [C] Trigonal bipyramidal, with bond angles of  $90^\circ$  and  $120^\circ$
- [D] Octahedral, with bond angles of  $90^\circ$

**73 [EASY]** Shape and Bond Angle of Sulfur Hexafluoride,  $\text{SF}_6$  — 9701/13/F/M/24/Q14**[1]**What is the geometry and F-S-F bond angle in sulfur hexafluoride,  $\text{SF}_6$ ?

- [A] Octahedral,  $90^\circ$
- [B] Hexagonal planar,  $60^\circ$
- [C] Trigonal bipyramidal,  $90^\circ$  and  $120^\circ$
- [D] Square planar,  $90^\circ$

**74 [EASY]** Shape and Bond Angle of the Sulfate Ion,  $\text{SO}_4^{2-}$  — 9701/11/M/J/22/Q17**[1]**What is the geometry and bond angle of the sulfate ion,  $\text{SO}_4^{2-}$ ?

- [A] Square planar,  $90^\circ$
- [B] Trigonal pyramidal,  $107^\circ$
- [C] Tetrahedral,  $109.5^\circ$
- [D] Octahedral,  $90^\circ$

**75 [HARD]** Shape of Chlorine Trifluoride,  $\text{ClF}_3$  — 9701/12/M/J/22/Q17**[1]**Chlorine trifluoride,  $\text{ClF}_3$ , has 5 electron pairs around the central chlorine atom (3 bonding pairs and 2 lone pairs).What is the molecular shape of  $\text{ClF}_3$ ?

- [A] Trigonal planar
- [B] T-shaped
- [C] Trigonal pyramidal
- [D] Tetrahedral

**76 [HARD]** Shape of Xenon Tetrafluoride,  $\text{XeF}_4$  — 9701/13/M/J/22/Q17**[1]**Xenon tetrafluoride,  $\text{XeF}_4$ , has six electron pairs around the central xenon atom (4 bonding pairs and 2 lone pairs).What is the shape and F-Xe-F bond angle of  $\text{XeF}_4$ ?

- [A] Tetrahedral,  $109.5^\circ$
- [B] Octahedral,  $90^\circ$
- [C] See-saw, approx  $88^\circ$  and  $118^\circ$
- [D] Square planar,  $90^\circ$

**77 [HARD]** Multiple Bond Angles in Organic Molecules: Ethanol — 9701/11/O/N/22/Q17**[1]**Consider a molecule of ethanol,  $\text{CH}_3\text{-CH}_2\text{-OH}$ .

What are the approximate bond angles around the methyl carbon (C1) and the oxygen atom (O)?

- [A] Around C1:  $120^\circ$ ; Around O:  $180^\circ$
- [B] Around C1:  $109.5^\circ$ ; Around O:  $120^\circ$
- [C] Around C1:  $109.5^\circ$ ; Around O:  $104.5^\circ$
- [D] Around C1:  $107^\circ$ ; Around O:  $109.5^\circ$

**78** [HARD] Bond Angle in Group 15 Hydrides:  $\text{NH}_3$  vs  $\text{PH}_3$  — 9701/12/O/N/22/Q17

[1]

The H-N-H bond angle in ammonia is  $107.0^\circ$ , whereas the H-P-H bond angle in phosphine,  $\text{PH}_3$ , is  $93.5^\circ$ .

Why is the bond angle in  $\text{PH}_3$  so much smaller than in  $\text{NH}_3$ ?

- [A] Phosphorus has a larger atomic radius and lower electronegativity than nitrogen, so P-H bonding pairs are further from the nucleus and experience less mutual repulsion.
- [B] Phosphorus possesses two non-bonding lone pairs while nitrogen possesses only one.
- [C] The P-H bonds are formed by unhybridized d orbitals rather than p orbitals.
- [D]  $\text{PH}_3$  has a square planar geometry while  $\text{NH}_3$  is tetrahedral.

**79** [EASY] Shape of the Nitronium Ion,  $\text{NO}_2^+$  — 9701/13/O/N/22/Q17

[1]

What is the shape and bond angle of the nitronium ion,  $\text{NO}_2^+$  (the electrophile in benzene nitration)?

- [A] Bent,  $104.5^\circ$
- [B] Trigonal planar,  $120^\circ$
- [C] Trigonal pyramidal,  $107^\circ$
- [D] Linear,  $180^\circ$

**80** [HARD] Shape of the Triiodide Ion,  $\text{I}_3^-$  — 9701/11/F/M/23/Q15

[1]

What is the molecular geometry of the triiodide ion,  $\text{I}_3^-$ ?

- [A] Bent ( $104.5^\circ$ )
- [B] Linear
- [C] Trigonal planar
- [D] T-shaped

**SUBTOPIC 3.6 — INTERMOLECULAR FORCES & HYDROGEN BONDING (Q81 – Q95)****81** [EASY] Origin of London Dispersion (Instantaneous Dipole-Induced Dipole) Forces — 9701/11/M/J/23/Q16

[1]

How do London dispersion forces (instantaneous dipole-induced dipole interactions) originate between non-polar atoms or molecules?

- [A] Permanent electronegativity differences create permanent partial charges that attract adjacent molecules.
- [B] Random fluctuations in the movement of electrons create a temporary instantaneous dipole, which polarises the electron cloud of a neighbouring molecule to induce a complementary dipole.
- [C] Hydrogen atoms bonded to electronegative atoms interact with lone pairs on adjacent molecules.
- [D] Covalent bonds continuously break and reform between adjacent molecules in the liquid state.

**82** [EASY] Factors Influencing London Dispersion Force Strength — 9701/12/M/J/23/Q16

[1]

Why does the boiling point of the noble gases increase steadily down Group 18:

$\text{He} (-269^\circ\text{C}) < \text{Ne} (-246^\circ\text{C}) < \text{Ar} (-186^\circ\text{C}) < \text{Kr} (-153^\circ\text{C}) < \text{Xe} (-108^\circ\text{C})$ ?

- [A] Nuclear charge increases, pulling noble gas atoms into permanent dipole configurations.
- [B] The atomic radius decreases, allowing atoms to pack closer together.
- [C] Intermolecular hydrogen bonding becomes progressively stronger down the group.
- [D] Atomic number and total number of electrons increase, making the electron cloud larger and more easily polarised, resulting in stronger London dispersion forces.

**83 [HARD]** Molecular Shape and London Dispersion Forces — 9701/13/M/J/23/Q16**[1]**

Pentane and 2,2-dimethylpropane both have the identical molecular formula  $C_5H_{12}$  ( $M_r = 72.0$ ). However, pentane boils at  $36.1\text{ }^{\circ}\text{C}$  while 2,2-dimethylpropane boils at  $9.5\text{ }^{\circ}\text{C}$ .

What accounts for this significant difference in boiling points?

- [A]** Pentane has an elongated, straight chain with a larger surface area of contact, enabling stronger London dispersion forces than the compact, spherical 2,2-dimethylpropane.
- [B]** Pentane contains permanent dipole-dipole attractions while 2,2-dimethylpropane is non-polar.
- [C]** 2,2-dimethylpropane has stronger intramolecular C-C covalent bonds that lower its volatility.
- [D]** Pentane forms intermolecular hydrogen bonds in the liquid state.

**84 [EASY]** Permanent Dipole-Dipole Forces — 9701/11/O/N/23/Q18**[1]**

Between which pair of molecules do permanent dipole-dipole forces act as the predominant intermolecular attraction?

- [A]** Methane,  $CH_4$ , and methane,  $CH_4$
- [B]** Water,  $H_2O$ , and water,  $H_2O$
- [C]** Propanone,  $CH_3COCH_3$ , and propanone,  $CH_3COCH_3$
- [D]** Helium,  $He$ , and helium,  $He$

**85 [EASY]** Essential Conditions for Hydrogen Bonding — 9701/12/O/N/23/Q18**[1]**

Which set of conditions is strictly required for the formation of an intermolecular hydrogen bond?

- [A]** A hydrogen atom covalently bonded to a highly electronegative atom (N, O, or F) and an unshared lone pair of electrons on an electronegative atom (N, O, or F) in an adjacent molecule.
- [B]** Any hydrogen atom bonded to any non-metal atom interacting with an electron cloud.
- [C]** A proton interacting with delocalised electrons in a metallic lattice.
- [D]** A covalent bond between hydrogen and chlorine in an isolated diatomic molecule.

**86 [HARD]** Anomalous Boiling Points of Period 2 Hydrides — 9701/13/O/N/23/Q18**[1]**

In Group 16 hydrides, the boiling points follow the trend:

$H_2O$  ( $100\text{ }^{\circ}\text{C}$ )  $\gg H_2S$  ( $-60\text{ }^{\circ}\text{C}$ )  $< H_2Se$  ( $-41\text{ }^{\circ}\text{C}$ )  $< H_2Te$  ( $-2\text{ }^{\circ}\text{C}$ )

Why does water have an extraordinarily high boiling point compared to hydrogen sulfide?

- [A]** The O-H covalent bond within a water molecule is much stronger than the S-H covalent bond.
- [B]** Water has a higher relative molecular mass than hydrogen sulfide.
- [C]** Water undergoes partial thermal dissociation into  $H^+$  and  $OH^-$  ions in the liquid state.
- [D]** Water molecules form extensive intermolecular hydrogen bonds, which are substantially stronger than the London dispersion and permanent dipole forces in  $H_2S$ .

**87 [HARD]** Boiling Point Comparison: Water vs Hydrogen Fluoride — 9701/11/F/M/24/Q15**[1]**

Fluorine is more electronegative than oxygen (3.98 vs 3.44), and individual H-F hydrogen bonds are stronger than individual O-H hydrogen bonds.

Why, then, does water (boiling point  $100\text{ }^{\circ}\text{C}$ ) boil at a significantly higher temperature than hydrogen fluoride (boiling point  $19.5\text{ }^{\circ}\text{C}$ )?

- [A]** Water has a greater relative molecular mass than hydrogen fluoride.
- [B]** Each water molecule can form an average of four hydrogen bonds, whereas each hydrogen fluoride molecule can form an average of only two hydrogen bonds.
- [C]** Hydrogen fluoride molecules dimerize into non-polar rings in the liquid state.
- [D]** Water has stronger London dispersion forces due to having more electrons.

**88 [HARD]** Anomalous Density of Ice Compared to Liquid Water — 9701/12/F/M/24/Q15**[1]**

Unlike almost all other substances, solid water (ice) is less dense than liquid water at 0 °C and floats.

What structural feature of ice explains this phenomenon?

- [A]** Ice contains trapped microscopic bubbles of air that lower its macroscopic density.
- [B]** The O–H covalent bonds lengthen significantly upon freezing, expanding the molecules.
- [C]** In ice, water molecules are held in an open, rigid, three-dimensional tetrahedral lattice by hydrogen bonds, creating large hexagonal cavities that collapse upon melting.
- [D]** Water molecules lose their permanent dipole moments when freezing into a crystalline lattice.

**89 [EASY]** Boiling Points of Isomeric Alcohols and Ethers — 9701/13/F/M/24/Q15**[1]**

Ethanol ( $\text{CH}_3\text{CH}_2\text{OH}$ ) and methoxymethane ( $\text{CH}_3\text{OCH}_3$ ) are functional group isomers with the identical molecular formula  $\text{C}_2\text{H}_6\text{O}$  ( $M_r = 46.0$ ).

Ethanol boils at 78.3 °C while methoxymethane boils at –24.8 °C. Why is the boiling point of ethanol so much higher?

- [A]** Ethanol has stronger covalent bonds than methoxymethane.
- [B]** Ethanol molecules form intermolecular hydrogen bonds via their –OH groups, whereas methoxymethane molecules cannot form hydrogen bonds with each other.
- [C]** Methoxymethane is completely non-polar and has zero dipole moment.
- [D]** Ethanol has a larger molecular surface area that doubles its London dispersion forces.

**90 [EASY]** Solubility of Alcohols in Water — 9701/11/M/J/22/Q18**[1]**

Why are low molecular mass alcohols such as methanol and ethanol completely miscible with water in all proportions?

- [A]** Alcohol molecules can form intermolecular hydrogen bonds with water molecules.
- [B]** Alcohols react chemically with water to form stable oxonium hydroxide salts.
- [C]** Alcohols have identical molecular geometry and bond angles to water.
- [D]** The non-polar hydrocarbon chains dissolve in the polar water structure via hydrophobic bonds.

**91 [HARD]** Decrease in Alcohol Solubility with Increasing Chain Length — 9701/12/M/J/22/Q18**[1]**

While ethanol is completely miscible with water, hexan-1-ol ( $\text{CH}_3(\text{CH}_2)_5\text{OH}$ ) is almost completely insoluble in water.

What is the explanation for this sharp decline in solubility?

- [A]** Hexan-1-ol loses its ability to form hydrogen bonds due to steric hindrance around the –OH group.
- [B]** The density of hexan-1-ol is greater than that of water, preventing mixing.
- [C]** Hexan-1-ol undergoes intramolecular hydrogen bonding that deactivates its hydroxyl group.
- [D]** The large non-polar alkyl chain is hydrophobic and disrupts the hydrogen-bonding network between water molecules without forming compensating attractions.

**92 [HARD]** Energy Scale: Intermolecular Forces vs Covalent Bonds — 9701/13/M/J/22/Q18**[1]**

When liquid water boils at 100 °C to form steam, approximately 40.7 kJ mol<sup>–1</sup> of energy is absorbed. However, breaking the O–H bonds in gaseous water into isolated atoms requires 928 kJ mol<sup>–1</sup>.

What fundamental conclusion does this comparison demonstrate?

- [A]** Boiling water involves the complete homolytic cleavage of covalent O–H bonds.
- [B]** Liquid water contains only covalent bonds, while steam contains only hydrogen bonds.
- [C]** Intermolecular hydrogen bonds are roughly an order of magnitude weaker than intramolecular covalent bonds.
- [D]** Hydrogen bonds require more energy to break than covalent bonds at high temperatures.

**93 [EASY]** Intermolecular Forces in Liquid Halogens — 9701/11/O/N/22/Q18**[1]**

At room temperature and pressure, chlorine is a gas, bromine is a volatile liquid, and iodine is a crystalline solid.

What accounts for this progressive change in physical state down Group 17?

- [A]** The strength of London dispersion forces increases as the total number of electrons and polarisability of the halogen molecules increase.
- [B]** The strength of the covalent halogen-halogen bond increases down the group.
- [C]** Electronegativity increases down Group 17, creating stronger dipole-dipole attractions.
- [D]** Iodine molecules form intermolecular coordinate bonds in the solid state.

**94 [HARD]** Hydrogen Bonding in Biological Macromolecules: DNA — 9701/12/O/N/22/Q18**[1]**

In the double helix structure of DNA, complementary nitrogenous bases pair across opposite strands:

Adenine pairs with Thymine via 2 hydrogen bonds;

Guanine pairs with Cytosine via 3 hydrogen bonds.

Why does a DNA fragment rich in Guanine-Cytosine (G-C) base pairs require a higher temperature to denature (separate) than one rich in Adenine-Thymine (A-T) pairs?

- [A]** G-C pairs form covalent cross-links between the two deoxyribose-phosphate backbones.
- [B]** Cytosine is more electronegative than thymine, creating an ionic bond with guanine.
- [C]** Three hydrogen bonds per G-C pair provide greater cumulative intermolecular stability than two hydrogen bonds per A-T pair.
- [D]** G-C pairs undergo metallic delocalisation along the vertical axis of the double helix.

**95 [HARD]** Viscosity Trend in Polyhydric Alcohols — 9701/13/O/N/22/Q18**[1]**

Propane-1,2,3-triol (glycerol) is an exceptionally viscous, syrupy liquid compared to propan-1-ol.

What is the primary cause of glycerol's high viscosity?

- [A]** Glycerol molecules form strong covalent cross-links with each other at room temperature.
- [B]** Each glycerol molecule has three hydroxyl (–OH) groups, establishing an extensive, highly tangled three-dimensional network of intermolecular hydrogen bonds.
- [C]** Glycerol has a cyclic structure that prevents molecular flow.
- [D]** The molecular mass of glycerol is over ten times greater than that of propan-1-ol.

**SUBTOPIC 3.7 — DOT-AND-CROSS DIAGRAMS & LEWIS STRUCTURES (Q96 – Q100)****96 [EASY]** Electron-Deficient Lewis Structure: Boron Trifluoride — 9701/11/M/J/23/Q17**[1]**

In the correct Lewis dot-and-cross diagram of a boron trifluoride, BF<sub>3</sub>, molecule, how many valence electrons surround the central boron atom?

- [A]** 8 valence electrons (a complete octet)
- [B]** 10 valence electrons (an expanded octet)
- [C]** 4 valence electrons (two bonding pairs)
- [D]** 6 valence electrons (an incomplete octet)

**97 [EASY]** Expanded Octet: Phosphorus Pentachloride — 9701/12/M/J/23/Q17**[1]**

Why can phosphorus form phosphorus pentachloride, PCl<sub>5</sub>, while nitrogen (in the same group) cannot form NCl<sub>5</sub>?

- [A]** Nitrogen is more electronegative than chlorine, which prevents covalent bonding.
- [B]** The N-Cl bond enthalpy is too high for multiple bonds to form.
- [C]** Phosphorus is in Period 3 and has energetically accessible, vacant 3d orbitals to expand its octet, whereas nitrogen is in Period 2 and has no d orbitals in its valence shell.
- [D]** Phosphorus has 10 valence electrons in its ground-state neutral atom.

**98** [EASY] Dot-and-Cross Diagram of Carbon Dioxide — 9701/13/M/J/23/Q17**[1]**

In the Lewis structure of carbon dioxide,  $\text{CO}_2$ , what is the total number of shared bonding electrons and non-bonding lone pairs in the entire molecule?

- [A] 8 shared bonding electrons (two double bonds) and 4 non-bonding lone pairs (two on each oxygen)
- [B] 4 shared bonding electrons and 8 non-bonding lone pairs
- [C] 8 shared bonding electrons and 0 non-bonding lone pairs
- [D] 6 shared bonding electrons and 6 non-bonding lone pairs

**99** [HARD] Dot-and-Cross Diagram of the Nitrite Ion,  $\text{NO}_2^-$  — 9701/11/O/N/23/Q19**[1]**

How many non-bonding lone pairs of electrons are present on the central nitrogen atom in the nitrite ion,  $\text{NO}_2^-$ ?

- [A] 0 lone pairs
- [B] 2 lone pairs
- [C] 3 lone pairs
- [D] 1 lone pair

**100** [HARD] Comprehensive Chemical Bonding Review — 9701/12/O/N/23/Q19**[1]**

Which statement regarding chemical bonding and molecular structure is FALSE?

- [A] Ionic compounds conduct electricity when molten because the ions are free to move.
- [B] A coordinate (dative covalent) bond is inherently weaker and longer than a normal covalent bond once it has formed.
- [C] Ice is less dense than liquid water due to an open hydrogen-bonded tetrahedral lattice.
- [D] Bond angles in VSEPR decrease in the order:  $\text{CH}_4$  ( $109.5^\circ$ ) >  $\text{NH}_3$  ( $107^\circ$ ) >  $\text{H}_2\text{O}$  ( $104.5^\circ$ ).

**FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)****101** [HARD] [HIGH-FREQUENCY EXAM REPEAT] Dative Covalent Bonding in Aluminium Chloride Dimer ( $\text{Al}_2\text{Cl}_6$ ) — 9701/11/M/J/23/Q18**[1]**

Aluminium chloride vapour at  $150^\circ\text{C}$  exists predominantly as  $\text{Al}_2\text{Cl}_6$  dimers.

How many coordinate (dative covalent) bonds are present in one molecule of  $\text{Al}_2\text{Cl}_6$ , and which atom acts as the electron pair donor?

- [A] 1 coordinate bond; aluminium acts as the electron pair donor.
- [B] 4 coordinate bonds; chlorine atoms act as electron pair donors.
- [C] 2 coordinate bonds; chlorine atoms act as electron pair donors.
- [D] 2 coordinate bonds; aluminium atoms act as electron pair donors.

**102** [EASY] [HIGH-FREQUENCY EXAM REPEAT] Equivalence of Bonds in the Ammonium Cation ( $\text{NH}_4^+$ ) — 9701/12/M/J/23/Q18**[1]**

The ammonium ion,  $\text{NH}_4^+$ , is formed when ammonia,  $\text{NH}_3$ , reacts with a hydrogen ion,  $\text{H}^+$ .

Which statement accurately describes the four N-H bonds in the resulting  $\text{NH}_4^+$  ion?

- [A] All four N-H bonds are identical in length, bond energy, and electron distribution once formed.
- [B] Three bonds are ordinary covalent bonds and one bond is a coordinate bond with distinct length and properties.
- [C] The bond formed by the  $\text{H}^+$  ion is purely electrostatic (ionic) while the other three are covalent.
- [D] Two bonds are sigma bonds and two bonds are pi bonds.

**103** [HARD] [HIGH-FREQUENCY EXAM REPEAT] VSEPR Repulsion Hierarchy and Bond Angle Progression — 9701/13/M/J/23/Q18**[1]**

Consider the three molecules: methane ( $\text{CH}_4$ ), ammonia ( $\text{NH}_3$ ), and water ( $\text{H}_2\text{O}$ ).

Why do the bond angles decrease in the order  $\text{CH}_4$  ( $109.5^\circ$ ) >  $\text{NH}_3$  ( $107.0^\circ$ ) >  $\text{H}_2\text{O}$  ( $104.5^\circ$ )?

- [A] The electronegativity of the central atom decreases from carbon to oxygen, weakening the bonds.
- [B] The atomic radius of the central atom increases from carbon to oxygen, forcing hydrogen atoms closer.
- [C] Methane has  $\text{sp}^3$  hybridization, whereas water has unhybridized p orbitals.
- [D] Lone pairs have greater repulsive force than bonding pairs; each additional lone pair compresses adjacent bonding pairs by approximately  $2.5^\circ$ .

**104** [HARD] [HIGH-FREQUENCY EXAM REPEAT] Octet Expansion and Hypervalency in Period 3 Elements — 9701/11/O/N/23/Q20**[1]**

Which molecule contains a central atom that has expanded its valence shell to accommodate more than eight electrons?

- [A] Carbon tetrachloride,  $\text{CCl}_4$  (8 valence electrons on C)
- [B] Sulfur hexafluoride,  $\text{SF}_6$  (12 valence electrons on S)
- [C] Nitrogen trifluoride,  $\text{NF}_3$  (8 valence electrons on N)
- [D] Boron trifluoride,  $\text{BF}_3$  (6 valence electrons on B)

**105** [HARD] [HIGH-FREQUENCY EXAM REPEAT] Anomalous Density and Hydrogen-Bonded Lattice of Ice — 9701/12/O/N/23/Q20**[1]**

Why is ice at  $0^\circ\text{C}$  less dense than liquid water at  $0^\circ\text{C}$ ?

- [A] Hydrogen bonds hold water molecules in a rigid, open three-dimensional tetrahedral lattice with relatively large empty spaces.
- [B] Ice contains covalent cross-links that push molecules further apart.
- [C] Water molecules in ice have higher kinetic energy and vibrate across larger amplitudes.
- [D] Liquid water contains dissolved gas bubbles that increase its density.

**106** [HARD] [HIGH-FREQUENCY EXAM REPEAT] Hydrogen Bonding Capacity: Water vs Hydrogen Fluoride — 9701/13/O/N/23/Q20**[1]**

Although the H-F bond is more polar than the O-H bond, water boils at  $100^\circ\text{C}$  while hydrogen fluoride boils at  $19.5^\circ\text{C}$ .

What is the fundamental scientific reason for water's higher boiling point?

- [A] Water has a higher relative molecular mass than hydrogen fluoride.
- [B] The O-H covalent bond is stronger than the H-F covalent bond.
- [C] Water has an average of four hydrogen bonds per molecule (2 H atoms and 2 lone pairs), whereas HF has an average of only two hydrogen bonds per molecule (limited by having only 1 H atom).
- [D] Liquid HF forms unreactive ring trimers that do not interact with other molecules.

**107** [HARD] [HIGH-FREQUENCY EXAM REPEAT] Counting Sigma and Pi Bonds in Unsaturated Molecules — 9701/11/F/M/24/Q16**[1]**

How many sigma ( $\sigma$ ) bonds and pi ( $\pi$ ) bonds are present in one molecule of propyne,  $\text{CH}_3\text{-C}\equiv\text{CH}$ ?

- [A] 5 sigma bonds and 3 pi bonds
- [B] 6 sigma bonds and 2 pi bonds
- [C] 7 sigma bonds and 1 pi bond
- [D] 6 sigma bonds and 1 pi bond

**108** [EASY] [HIGH-FREQUENCY EXAM REPEAT] Symmetrical Cancellation of Bond Dipoles — 9701/12/F/M/24/Q16**[1]**

Which molecule contains polar covalent bonds but has zero resultant molecular dipole moment due to spatial symmetry?

- [A] Water,  $\text{H}_2\text{O}$
- [B] Ammonia,  $\text{NH}_3$
- [C] Sulfur dioxide,  $\text{SO}_2$
- [D] Tetrachloromethane,  $\text{CCl}_4$

**109** [HARD] [HIGH-FREQUENCY EXAM REPEAT] Lattice Energy Magnitude and Melting Points of Giant Ionic Solids — 9701/13/F/M/24/Q16

[1]

The melting point of magnesium oxide, MgO, is 2852 °C, whereas the melting point of sodium chloride, NaCl, is 801 °C.

Which statement correctly explains this enormous difference in melting points?

- [A]  $\text{Mg}^{2+}$  and  $\text{O}^{2-}$  ions have double the charges (+2 and -2) and smaller ionic radii than  $\text{Na}^{+}$  and  $\text{Cl}^{-}$  ions, producing vastly stronger electrostatic attraction.
- [B] MgO forms a giant covalent macromolecular network while NaCl is giant ionic.
- [C] Magnesium has a higher electronegativity than sodium, causing MgO to be covalent.
- [D] NaCl has weak intermolecular forces holding ion pairs together.

**110** [HARD] [HIGH-FREQUENCY EXAM REPEAT] Energy Scale: Intermolecular Forces vs Intramolecular Covalent Bonds — 9701/11/M/J/22/Q20

[1]

Which statement regarding the relative energies of intermolecular forces and covalent bonds is correct?

- [A] Hydrogen bonds are stronger than single covalent bonds but weaker than double covalent bonds.
- [B] London dispersion forces require more thermal energy to disrupt than ionic lattice bonds.
- [C] Overcoming intermolecular forces during the boiling of liquid water requires  $\sim 41 \text{ kJ mol}^{-1}$ , whereas breaking the intramolecular covalent O-H bonds requires  $\sim 460 \text{ kJ mol}^{-1}$ .
- [D] Boiling a molecular liquid involves breaking intramolecular covalent bonds into gaseous atoms.

### QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

| Q1 – Q20 |   | Q21 – Q40 |   | Q41 – Q60 |   | Q61 – Q80 |   | Q81 – Q100 |   |
|----------|---|-----------|---|-----------|---|-----------|---|------------|---|
| Q01      | B | Q21       | B | Q41       | B | Q61       | B | Q81        | B |
| Q02      | D | Q22       | D | Q42       | D | Q62       | D | Q82        | D |
| Q03      | A | Q23       | A | Q43       | A | Q63       | A | Q83        | A |
| Q04      | C | Q24       | C | Q44       | C | Q64       | C | Q84        | C |
| Q05      | A | Q25       | A | Q45       | A | Q65       | A | Q85        | A |
| Q06      | D | Q26       | D | Q46       | D | Q66       | D | Q86        | D |
| Q07      | B | Q27       | B | Q47       | B | Q67       | B | Q87        | B |
| Q08      | C | Q28       | C | Q48       | C | Q68       | C | Q88        | C |
| Q09      | B | Q29       | B | Q49       | B | Q69       | B | Q89        | B |
| Q10      | A | Q30       | A | Q50       | A | Q70       | A | Q90        | A |
| Q11      | D | Q31       | D | Q51       | D | Q71       | D | Q91        | D |
| Q12      | C | Q32       | C | Q52       | C | Q72       | C | Q92        | C |
| Q13      | A | Q33       | A | Q53       | A | Q73       | A | Q93        | A |
| Q14      | C | Q34       | C | Q54       | C | Q74       | C | Q94        | C |
| Q15      | B | Q35       | B | Q55       | B | Q75       | B | Q95        | B |
| Q16      | D | Q36       | D | Q56       | D | Q76       | D | Q96        | D |
| Q17      | C | Q37       | C | Q57       | C | Q77       | C | Q97        | C |
| Q18      | A | Q38       | A | Q58       | A | Q78       | A | Q98        | A |
| Q19      | D | Q39       | D | Q59       | D | Q79       | D | Q99        | D |
| Q20      | B | Q40       | B | Q60       | B | Q80       | B | Q100       | B |

### FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

|           |           |           |           |           |           |           |           |           |           |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Q101<br>C | Q102<br>A | Q103<br>D | Q104<br>B | Q105<br>A | Q106<br>C | Q107<br>B | Q108<br>D | Q109<br>A | Q110<br>C |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|

COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

| Q  | KEY | EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | MARKS |
|----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1  | B   | Option B is the correct and standard IUPAC/Cambridge definition of electronegativity. Option A defines first electron affinity. Option C defines metallic bonding. Option D defines first ionisation energy.                                                                                                                                                                                                                                                                                                                                                                         | [1]   |
| 2  | D   | Option D is correct. Across Period 3, proton number increases from 11 (Na) to 17 (Cl) while electrons enter the same third principal quantum shell (shielding remains roughly constant). Consequently, effective nuclear charge increases and atomic radius decreases, enabling the nucleus to exert a substantially stronger electrostatic pull on bonded electron pairs.                                                                                                                                                                                                           | [1]   |
| 3  | A   | Option A is correct. Descending Group 17, atomic radius increases as additional electron shells are added, and inner shell shielding increases. Although nuclear charge increases, the increased distance and shielding dominate, reducing the electrostatic pull on shared electron pairs. Fluorine is the most electronegative element (Pauling value ~4.0), followed by Cl (3.0), Br (2.8), and I (2.5).                                                                                                                                                                          | [1]   |
| 4  | C   | Option C is correct. In CCl <sub>4</sub> , each C–Cl bond is polar due to the electronegativity difference between C (2.5) and Cl (3.0). However, CCl <sub>4</sub> has a perfectly symmetrical regular tetrahedral geometry (bond angles 109.5°). The four individual C–Cl bond dipole vectors cancel each other out completely in three-dimensional space, yielding a resultant molecular dipole moment of zero. In CHCl <sub>3</sub> , CH <sub>2</sub> Cl <sub>2</sub> , and CH <sub>3</sub> Cl, the bond dipoles do not cancel symmetrically, resulting in net molecular dipoles. | [1]   |
| 5  | A   | Option A is correct. CO <sub>2</sub> has 2 bonding regions and 0 lone pairs on carbon, giving a linear geometry (180°); the two C=O dipole vectors point in exactly opposite directions and cancel completely. SO <sub>2</sub> has 2 bonding regions and 1 lone pair on sulfur, giving a bent/non-linear geometry (~119°); the two S=O dipoles do not cancel and result in a permanent net dipole moment.                                                                                                                                                                            | [1]   |
| 6  | D   | Option D is correct. The degree of ionic character is directly proportional to the difference in electronegativity (Delta EN) between the bonded atoms. For Cs–F, Delta EN = 3.98 – 0.79 = 3.19. For Na–F, Delta EN = 3.98 – 0.93 = 3.05. For Na–Cl, Delta EN = 3.16 – 0.93 = 2.23. For H–F, Delta EN = 3.98 – 2.20 = 1.78. Cs–F exhibits the greatest electronegativity difference, imparting the highest percent ionic character.                                                                                                                                                  | [1]   |
| 7  | B   | Option B is correct. According to Fajan's rules, covalent character in an ionic compound is maximized by: (1) a small, highly charged cation with high charge density (polarising power), and (2) a large, easily distorted anion with low charge density (polarisability). Al <sup>3+</sup> has a high charge (+3) and small ionic radius (53 pm), while I <sup>–</sup> has a large electron cloud (220 pm) that is readily pulled towards the Al <sup>3+</sup> cation, leading to substantial orbital overlap and covalent character.                                              | [1]   |
| 8  | C   | Option C is correct. BF <sub>3</sub> has 3 bonding pairs and 0 lone pairs around the central boron atom, giving a trigonal planar geometry with 120° bond angles. The vector sum of three equal coplanar dipoles pointing towards the vertices of an equilateral triangle is mathematically zero.                                                                                                                                                                                                                                                                                    | [1]   |
| 9  | B   | Option B is correct. In CH <sub>3</sub> Cl, the single C–Cl dipole is reinforced by the small C–H bond dipoles pointing towards carbon, yielding a large resultant dipole moment (~1.87 D). In CH <sub>2</sub> Cl <sub>2</sub> (~1.60 D) and CHCl <sub>3</sub> (~1.04 D), the multiple C–Cl dipoles point partially away from each other, reducing the net vector sum. In CCl <sub>4</sub> , complete tetrahedral symmetry reduces the net dipole to 0 D.                                                                                                                            | [1]   |
| 10 | A   | Option A is correct. Be <sup>2+</sup> has a minute ionic radius (31 pm) compared to Ba <sup>2+</sup> (135 pm). With a +2 charge concentrated over a tiny sphere, Be <sup>2+</sup> possesses an exceptionally high charge density and polarising power. When interacting with Cl <sup>–</sup> anions, Be <sup>2+</sup> distorts their electron clouds so intensely that electron sharing (covalent bonding) occurs. Ba <sup>2+</sup> has very low charge density, leading to pure ionic bonding.                                                                                      | [1]   |
| 11 | D   | Option D is correct. The large electronegativity difference in HF (1.78) shortens the bond and creates strong ionic resonance energy, resulting in a very high H–F bond enthalpy (562 kJ/mol). As halogen electronegativity decreases and atomic radius increases down Group 17, bond length increases and bond enthalpy drops sharply (HCl: 431, HBr: 366, HI: 299 kJ/mol). Consequently, thermal stability decreases down the group.                                                                                                                                               | [1]   |
| 12 | C   | Option C is correct. Electronegativity difference in N–H is 3.04 – 2.20 = 0.84 (strongly polar). In P–H, the difference is  2.19 – 2.20  = 0.01 (essentially non-polar). Thus N–H bonds are far more polar than P–H bonds.                                                                                                                                                                                                                                                                                                                                                           | [1]   |
| 13 | A   | Option A is correct. In trans-1,2-dichloroethene, the two C–Cl bond dipoles point in equal and opposite directions across the center of inversion, cancelling each other out completely (net dipole = 0 D). In cis-1,2-dichloroethene, both C–Cl dipoles point toward the same side of the molecule, reinforcing each other to produce a permanent net dipole moment (~1.90 D).                                                                                                                                                                                                      | [1]   |
| 14 | C   | Option C is correct. Oxidation states are assigned by formally assigning bonded electrons to the more electronegative atom. Since fluorine (3.98) is more electronegative than oxygen (3.44), each F atom is assigned an oxidation state of –1. To maintain electrical neutrality, oxygen must be in the +2 oxidation state.                                                                                                                                                                                                                                                         | [1]   |
| 15 | B   | Option B is the accurate definition of ionic bonding. Ionic bonds are non-directional electrostatic forces of attraction that act uniformly in all directions between positive and negative ions, building a continuous giant lattice. Option A describes covalent bonding. Option C describes metallic bonding. Option D describes permanent dipole-dipole forces.                                                                                                                                                                                                                  | [1]   |
| 16 | D   | Option D is correct. Sodium chloride crystallizes in a face-centred cubic lattice where each Na <sup>+</sup> cation is octahedrally surrounded by 6 nearest-neighbour Cl <sup>–</sup> anions, and each Cl <sup>–</sup> anion is surrounded by 6 nearest-neighbour Na <sup>+</sup> cations. This is described as a 6:6 coordination lattice.                                                                                                                                                                                                                                          | [1]   |
| 17 | C   | Option C is correct. According to Coulomb's law, electrostatic lattice energy is proportional to (q <sub>1</sub> × q <sub>2</sub> ) / (r <sub>+</sub> + r <sub>–</sub> ), where q <sub>1</sub> and q <sub>2</sub> are ionic charges and (r <sub>+</sub> + r <sub>–</sub> ) is the sum of the ionic radii. Higher charges increase the numerator, while smaller radii decrease the inter-ionic separation in the denominator, both maximizing electrostatic attraction and making lattice enthalpy substantially more exothermic.                                                     | [1]   |

| Q  | KEY | EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | MARKS |
|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 18 | A   | Option A is correct. In MgO, the ions are Mg <sup>2+</sup> (radius 72 pm) and O <sup>2-</sup> (radius 140 pm), carrying 2+ and 2- charges (charge product = 4). In NaCl, the ions are Na <sup>+</sup> (102 pm) and Cl <sup>-</sup> (181 pm), carrying 1+ and 1- charges (charge product = 1). The fourfold increase in charge product combined with smaller inter-ionic distance makes the lattice enthalpy of MgO (-3791 kJ/mol) nearly five times greater than that of NaCl (-787 kJ/mol), requiring far higher thermal energy to overcome. | [1]   |
| 19 | D   | Option D is correct. In the solid state, K <sup>+</sup> and Br <sup>-</sup> ions are rigidly locked in fixed positions within the giant lattice and cannot migrate under an applied electric field (insulator). When molten (liquid) or dissolved in water, the lattice is broken down, freeing the ions to move towards the oppositely charged electrodes and carry electric current.                                                                                                                                                        | [1]   |
| 20 | B   | Option B is correct. When an external mechanical shearing stress is applied to an ionic lattice, adjacent planes of ions slide past each other. This displacement brings ions of identical charge (e.g. Ca <sup>2+</sup> next to Ca <sup>2+</sup> , and F <sup>-</sup> next to F <sup>-</sup> ) into close proximity. The resulting sudden, massive electrostatic repulsion between like charges forces the lattice planes apart, causing catastrophic fracture along clean cleavage planes.                                                  | [1]   |
| 21 | B   | Option B is correct. Descending Group 2, the cation charge remains constant (+2) while the ionic radius increases steadily: Mg <sup>2+</sup> (72 pm) < Ca <sup>2+</sup> (100 pm) < Sr <sup>2+</sup> (118 pm) < Ba <sup>2+</sup> (135 pm). As inter-ionic separation increases, the electrostatic attraction between the metal cation and O <sup>2-</sup> weakens, decreasing the magnitude of the lattice enthalpy and lowering the melting point from MgO (2852 °C) down to BaO (1923 °C).                                                   | [1]   |
| 22 | D   | Option D is correct. Dissolution involves two competing energetic processes: breaking the giant ionic lattice (endothermic, +ΔH <sub>lattice</sub> ) and surrounding the gaseous ions with polar water molecules to form ion-dipole attractions (exothermic, ΔH <sub>hydration</sub> ). If the exothermic hydration enthalpies are sufficiently high, ΔH <sub>solution</sub> will be favorable, enabling dissolution assisted by an increase in entropy.                                                                                      | [1]   |
| 23 | A   | Option A is correct. All four compounds contain the Na <sup>+</sup> cation. The halide anions increase in size down Group 17: F <sup>-</sup> (133 pm) < Cl <sup>-</sup> (181 pm) < Br <sup>-</sup> (196 pm) < I <sup>-</sup> (220 pm). Because F <sup>-</sup> has the smallest radius, the inter-ionic distance in NaF is minimized, generating the strongest electrostatic attraction and the most exothermic lattice enthalpy (-918 kJ/mol for NaF vs -705 kJ/mol for NaI).                                                                 | [1]   |
| 24 | C   | Option C is correct. High melting point (772 °C), solid non-conductor, aqueous conductor, and solubility in water are classic hallmarks of a giant ionic lattice such as CaCl <sub>2</sub> . SiO <sub>2</sub> has an extremely high mp (~1710 °C) but is completely insoluble and does not conduct in water. Sucrose is a simple molecular solid with a low melting point (186 °C) and non-conducting solution. Graphite is an electrical conductor in the solid state and insoluble in water.                                                | [1]   |
| 25 | A   | Option A is correct. Because the Cs <sup>+</sup> cation (radius 167 pm) is considerably larger than Na <sup>+</sup> (102 pm), it can accommodate 8 surrounding Cl <sup>-</sup> anions around it without anion-anion steric clash. Each Cs <sup>+</sup> ion sits at the centre of a cube surrounded by 8 Cl <sup>-</sup> ions at the corners, and each Cl <sup>-</sup> ion is likewise surrounded by 8 Cs <sup>+</sup> ions, establishing an 8:8 coordination lattice.                                                                         | [1]   |
| 26 | D   | Option D is correct. X-ray crystallography maps the distribution of electron density in a crystal. For ionic compounds like NaCl, the contour lines form concentric spheres tightly localized around the atomic nuclei, and the electron density between adjacent nuclei drops effectively to zero. In covalent compounds, by contrast, contour lines show continuous high electron density ridges (electron sharing) bridging the bonded atoms.                                                                                              | [1]   |
| 27 | B   | Option B is correct. Group 2 metals form M <sup>2+</sup> cations by losing 2 valence electrons. Group 15 non-metals form X <sup>3-</sup> anions by gaining 3 electrons to complete their octet. To achieve electrical neutrality, the total positive charge must equal the total negative charge: 3 x (+2) + 2 x (-3) = 0. Therefore, the empirical formula is M <sub>3</sub> X <sub>2</sub> .                                                                                                                                                | [1]   |
| 28 | C   | Option C is correct. Potassium exists as K <sup>+</sup> cations. Since there are 4 K <sup>+</sup> cations, the complex anion must carry a charge of 4-, i.e., [Fe(CN) <sub>6</sub> ] <sup>4-</sup> . Cyanide is a uni-negative ligand (CN <sup>-</sup> ). Let the oxidation state of iron be x: x + 6(-1) = -4 → x - 6 = -4 → x = +2.                                                                                                                                                                                                         | [1]   |
| 29 | B   | Option B is the standard definition of metallic bonding. In a metal, the valence electrons detach from individual parent atoms and become delocalised throughout the entire crystal structure, creating an array of positive cations embedded in an omnipresent mobile electron cloud.                                                                                                                                                                                                                                                        | [1]   |
| 30 | A   | Option A is correct. Metallic bond strength depends on charge density and electron pool: Na forms Na <sup>+</sup> (releasing 1 e <sup>-</sup> per atom; radius 102 pm); Mg forms Mg <sup>2+</sup> (releasing 2 e <sup>-</sup> per atom; radius 72 pm); Al forms Al <sup>3+</sup> (releasing 3 e <sup>-</sup> per atom; radius 53 pm). Higher cation charge, smaller cation radius, and higher delocalised electron density collectively produce vastly stronger electrostatic attraction, raising the melting point.                          | [1]   |
| 31 | D   | Option D is correct. In metallic bonding, valence electrons do not belong to any single atom; they form a mobile delocalised conduction band. When an electric potential is applied across the metal, these mobile electrons drift towards the positive terminal, establishing an electric current in both solid and liquid phases.                                                                                                                                                                                                           | [1]   |
| 32 | C   | Option C is correct. Unlike semiconductors, metals experience increased electrical resistance as temperature rises. Higher thermal kinetic energy causes the fixed metal cations in the lattice to vibrate with greater amplitude around their equilibrium positions. These vigorous vibrations increase the frequency of collisions with drifting delocalised electrons (electron scattering), impeding electron flow and lowering conductivity.                                                                                             | [1]   |
| 33 | A   | Option A is correct. Because metallic bonds are non-directional, when an external shear force is applied, planes of positive cations can roll or slide over adjacent layers. The sea of delocalised electrons instantly adjusts to the new cation positions, maintaining cohesive electrostatic attraction and preventing fracture.                                                                                                                                                                                                           | [1]   |
| 34 | C   | Option C is correct. Pure copper consists of identical sized atoms arranged in regular close-packed layers that slide readily over each other. Introducing zinc atoms (which have a different radius) distorts the regular crystalline lattice layers. This structural irregularity creates 'pinning' effects that make it far more difficult for planes of atoms to slide, rendering the alloy much harder and stronger.                                                                                                                     | [1]   |

| Q  | KEY | EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED                                                                                                                                                                                                                                                                                                                                                                                                                                                             | MARKS |
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| 35 | B   | Option B is correct. Each Group 1 metal contributes exactly 1 electron per atom to the delocalised sea, forming singly charged M <sup>+</sup> cations. Descending the group, the ionic radius increases (Li <sup>+</sup> : 76 pm → Cs <sup>+</sup> : 167 pm). The increased distance between the positive cation nucleus and the delocalised electrons significantly weakens electrostatic attraction, resulting in progressively lower melting points.                                                        | [1]   |
| 36 | D   | Option D is correct. Thermal energy in metals is transferred predominantly by fast-moving delocalised electrons colliding with cooler lattice sites and electrons, transporting kinetic energy through the metal at high speed. Additionally, vibrations of adjacent cations (phonons) propagate heat through the lattice.                                                                                                                                                                                     | [1]   |
| 37 | C   | Option C is correct. Aluminium is a metal with a high density of delocalised conduction electrons (excellent conductor, $\sim 3.8 \times 10^{27}$ S/m). Silicon is a metalloid/semiconductor with a small band gap that allows slight electrical conduction at room temperature ( $\sim 1.5 \times 10^{-3}$ S/m). Sulfur is a simple molecular non-metal (S <sub>8</sub> molecules) with all electrons localized in covalent bonds and lone pairs, behaving as an electrical insulator ( $\sim 10^{-15}$ S/m). | [1]   |
| 38 | A   | Option A is correct. In transition metals, the 3d and 4s subshells are close in energy. Both 4s and partially filled 3d electrons can participate in delocalisation and metallic bonding. With many more delocalised electrons per atom (e.g. up to 5 or 6) interacting with smaller, highly charged cations, the metallic bonding is exceptionally strong, leading to high melting points (e.g. Fe: 1538 °C) and high tensile strength.                                                                       | [1]   |
| 39 | D   | Option D is correct. Due to strong relativistic contraction of the 6s orbital in heavy elements (lanthanide contraction + high Z), the 6s <sup>2</sup> pair of electrons in mercury is drawn close to the nucleus and held very tightly (inert pair behavior). These electrons participate weakly in metallic bonding, resulting in unusually weak cohesion between Hg atoms and a liquid state at room temperature.                                                                                           | [1]   |
| 40 | B   | Option B is correct. In both face-centred cubic (cubic close-packed, ccp) and hexagonal close-packed (hcp) crystal structures, each spherical metal atom is in direct contact with 12 nearest neighbours: 6 in its own layer, 3 in the layer above, and 3 in the layer below. This achieves the maximum packing efficiency for spheres of 74%.                                                                                                                                                                 | [1]   |
| 41 | B   | Option B is correct. In molten sodium chloride (an ionic substance), the charge carriers are mobile cations (Na <sup>+</sup> ) and anions (Cl <sup>-</sup> ) that migrate toward opposite electrodes where electrolysis occurs. In liquid aluminium (a molten metal), metallic bonding persists in the fluid state, and conduction occurs via the rapid drift of mobile delocalised electrons without chemical decomposition.                                                                                  | [1]   |
| 42 | D   | Option D is correct. The photoelectric effect demonstrates that valence electrons in metals occupy delocalised energy levels at the Fermi surface. When incident photons possess energy ( $h\nu$ ) greater than the metal's work function ( $\Phi$ ), the photon energy is absorbed by an electron, ejecting it from the metal surface.                                                                                                                                                                        | [1]   |
| 43 | A   | Option A is the formal definition of a covalent bond: an electrostatic attraction between two positively charged atomic nuclei and the negatively charged shared pair of electrons located in the region of orbital overlap between them.                                                                                                                                                                                                                                                                      | [1]   |
| 44 | C   | Option C is correct. A sigma bond is formed by the direct, end-to-end (head-on) coaxial overlap of atomic orbitals (such as s-s, s-p, or hybridized sp <sup>3</sup> -sp <sup>3</sup> , sp <sup>2</sup> -sp <sup>2</sup> ) along the imaginary line connecting the two nuclei (the internuclear axis). Electron density is concentrated symmetrically along this axis.                                                                                                                                          | [1]   |
| 45 | A   | Option A is correct. A pi bond is formed by the sideways (lateral) overlap of two adjacent, parallel, unhybridized p atomic orbitals (e.g. 2p <sub>z</sub> -2p <sub>z</sub> ). This produces two lobes of electron density located above and below the internuclear axis, with a nodal plane along the axis itself.                                                                                                                                                                                            | [1]   |
| 46 | D   | Option D is correct. Direct head-on orbital overlap along the internuclear axis provides maximum spatial overlap between atomic wavefunctions, resulting in a strong sigma bond (348 kJ/mol). Sideways overlap of p orbitals occurs further from the internuclear axis and is geometrically less effective (less orbital overlap volume). Thus, the pi bond enthalpy is 612 - 348 = 264 kJ/mol, which is substantially weaker than the sigma bond.                                                             | [1]   |
| 47 | B   | Option B is correct. The structural formula of ethyne is H-C#C-H. Each single C-H bond is a sigma bond (2 C-H sigma bonds). The C#C triple bond consists of one strong coaxial sigma bond and two mutually perpendicular sideways pi bonds. Total: 2 + 1 = 3 sigma bonds, and 2 pi bonds.                                                                                                                                                                                                                      | [1]   |
| 48 | C   | Option C is correct. Let us count every bond: Two C-H single bonds in CH <sub>2</sub> (2 sigma); One C=C double bond (1 sigma, 1 pi); One C-H single bond on central CH (1 sigma); One C-C single bond (1 sigma); One C#N triple bond (1 sigma, 2 pi). Total sigma bonds = 2 + 1 + 1 + 1 + 1 = 6 sigma bonds. Total pi bonds = 1 (from C=C) + 2 (from C#N) = 3 pi bonds.                                                                                                                                       | [1]   |
| 49 | B   | Option B is correct. As bond order increases from single (1) to double (2) to triple (3), more electron pairs are shared between the two nitrogen nuclei. The greater electrostatic attraction pulls the nuclei closer together (decreasing bond length: N-N = 145 pm, N=N = 125 pm, N#N = 110 pm) and requires vastly more energy to dissociate (increasing bond enthalpy: N-N = 163 kJ/mol, N=N = 409 kJ/mol, N#N = 945 kJ/mol).                                                                             | [1]   |
| 50 | A   | Option A is correct. A coordinate (or dative covalent) bond is formed when one atom (the donor, possessing a non-bonding lone pair) shares its pair of electrons with another atom or ion (the acceptor, possessing a vacant orbital).                                                                                                                                                                                                                                                                         | [1]   |
| 51 | D   | Option D is correct. Ammonia has three N-H covalent bonds and one lone pair on nitrogen. The lone pair is donated into the empty 1s orbital of an H <sup>+</sup> ion, forming a coordinate bond. Once formed, all four bonding pairs are completely indistinguishable; the positive charge is delocalised over the entire tetrahedral ion, and all four N-H bonds have identical bond lengths (103 pm) and bond energies (391 kJ/mol).                                                                         | [1]   |
| 52 | C   | Option C is correct. A water molecule has two O-H covalent bonds and two lone pairs on oxygen. When an acid dissociates in water, H <sup>+</sup> (which has a vacant 1s orbital) accepts a lone pair from oxygen, creating a dative covalent bond. The resulting H <sub>3</sub> O <sup>+</sup> ion has trigonal pyramidal geometry with one remaining lone pair.                                                                                                                                               | [1]   |

| Q  | KEY | EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | MARKS |
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| 53 | A   | Option A is correct. In monomeric $\text{AlCl}_3$ , aluminium has only 6 valence electrons (electron-deficient). To complete its octet, two $\text{AlCl}_3$ molecules dimerize: a non-bonding lone pair from a chlorine atom on one $\text{AlCl}_3$ coordinates into the empty 3p orbital of the aluminium atom on the other $\text{AlCl}_3$ . With two such dative bridging bonds, both aluminium atoms attain a stable octet of 8 electrons in a tetrahedral arrangement.                                                                                        | [1]   |
| 54 | C   | Option C is correct. Carbon has 4 valence electrons and oxygen has 6. They share two pairs of electrons in standard covalent fashion (one sigma, one pi). However, this leaves carbon with only 6 outer electrons. Oxygen then donates one of its non-bonding lone pairs into a vacant 2p orbital on carbon, forming a third (dative) pi bond ( $\text{C} \leftarrow \text{O}$ ). Both atoms thus achieve an octet.                                                                                                                                                | [1]   |
| 55 | B   | Option B is correct. In $\text{BF}_3$ , boron has 3 bonding pairs and 0 lone pairs (trigonal planar, $120^\circ$ bond angles, $\text{sp}^2$ hybridized). When nitrogen in $\text{NH}_3$ donates its lone pair into the empty 2p orbital of boron, a dative B-N bond is formed. Boron now has 4 bonding pairs and 0 lone pairs, changing its geometry to tetrahedral with bond angles close to $109.5^\circ$ ( $\text{sp}^3$ hybridized).                                                                                                                           | [1]   |
| 56 | D   | Option D is correct. Nucleophilic substitution reactivity of haloalkanes is governed primarily by bond enthalpy rather than bond polarity. Because iodine has the largest atomic radius, the C-I bond is the longest and has the smallest bond enthalpy (240 kJ/mol). It requires the least activation energy to break, making 1-iodobutane the most reactive.                                                                                                                                                                                                     | [1]   |
| 57 | C   | Option C is correct. Homolytic fission involves the symmetrical cleavage of a covalent bond where each atom leaves with one electron from the shared pair, producing uncharged species with unpaired electrons known as free radicals ( $\text{Cl}-\text{Cl} \rightarrow 2 \text{Cl}^\bullet$ ). Heterolytic fission (Option A) produces ions.                                                                                                                                                                                                                     | [1]   |
| 58 | A   | Option A is correct. A C-C single bond is a cylindrical coaxial sigma bond; rotating around this axis maintains full orbital overlap, allowing free rotation. In a C=C double bond, the pi bond is formed by parallel sideways overlap of p orbitals. Rotating one end of the bond by $90^\circ$ would destroy this parallel alignment and break the pi bond, requiring ~260 kJ/mol of energy. At room temperature, thermal energy is insufficient, leading to restricted rotation and geometric isomerism.                                                        | [1]   |
| 59 | D   | Option D is correct. In $\text{NO}_3^-$ , unhybridized 2p orbitals on the central nitrogen and the three surrounding oxygen atoms overlap sideways to form a continuous delocalised pi system containing 4 electrons spread over all four atoms. This delocalisation gives each N-O bond identical properties with a fractional bond order of $4/3 = 1.33$ and equal intermediate bond lengths.                                                                                                                                                                    | [1]   |
| 60 | B   | Option B is correct. In transition metal complex chemistry, neutral molecules or anions with lone pairs (ligands) act as Lewis bases by donating a lone pair of electrons into vacant hybrid $\text{d}^2\text{sp}^3$ or $\text{sp}^3\text{d}^2$ orbitals of the central metal cation (Lewis acid), establishing dative covalent (coordinate) bonds.                                                                                                                                                                                                                | [1]   |
| 61 | B   | Option B is correct. The $\text{N}\equiv\text{N}$ molecule is held together by a triple bond consisting of one strong sigma bond and two pi bonds. Breaking this triple bond requires an exceptionally high activation energy (bond dissociation energy = 945 kJ/mol), making molecular nitrogen chemically inert under normal conditions.                                                                                                                                                                                                                         | [1]   |
| 62 | D   | Option D is correct. In monomeric $\text{AlCl}_3$ , aluminium has 3 bonding pairs and 0 lone pairs, giving a trigonal planar shape with $120^\circ$ bond angles. In $\text{AlCl}_4^-$ , an extra $\text{Cl}^-$ ion donates a lone pair into the empty 3p orbital of Al, forming a coordinate bond. Now Al has 4 bonding pairs and 0 lone pairs, adopting a regular tetrahedral geometry with $109.5^\circ$ bond angles.                                                                                                                                            | [1]   |
| 63 | A   | Option A is correct. The strength of a polar covalent bond (e.g. H-F or C-F) is enhanced by the electrostatic attraction between the partial charges $\delta^+$ and $\delta^-$ on the bonded atoms (ionic resonance stabilization). This additional ionic character strengthens the bond above the purely covalent value.                                                                                                                                                                                                                                          | [1]   |
| 64 | C   | Option C is correct. In ethene ( $\text{H}_2\text{C}=\text{CH}_2$ ), each carbon atom forms 3 sigma bonds (two to H, one to C) using three equivalent $\text{sp}^2$ hybrid orbitals directed towards the corners of an equilateral triangle (trigonal planar, $\sim 120^\circ$ angles). The remaining unhybridized 2p orbital on each carbon overlaps sideways to form the pi bond.                                                                                                                                                                                | [1]   |
| 65 | A   | Option A is the fundamental tenet of VSEPR theory: valence electron pairs (both bonding pairs and non-bonding lone pairs) surrounding a central atom are negatively charged regions that repel one another. To minimize overall electrostatic repulsion and maximize stability, these electron pairs arrange themselves as far apart in three-dimensional space as geometrically possible.                                                                                                                                                                         | [1]   |
| 66 | D   | Option D is correct. Non-bonding lone pairs are held by only ONE positive atomic nucleus, so their electron clouds are concentrated closer to the central atom and spread out wider in space. Bonding pairs are held between TWO nuclei and are more localized/confined. Therefore, lone pair - lone pair repulsion is greatest, followed by lone pair - bonding pair, with bonding pair - bonding pair being the weakest.                                                                                                                                         | [1]   |
| 67 | B   | Option B is correct. All three molecules have a tetrahedral electron-pair arrangement (4 pairs). $\text{CH}_4$ has 4 bonding pairs and 0 lone pairs (regular tetrahedron, $109.5^\circ$ ). $\text{NH}_3$ has 3 bonding pairs and 1 lone pair; the extra repulsion from the lone pair compresses the H-N-H bond angles to $\sim 107^\circ$ ( $\sim 2.5^\circ$ reduction). $\text{H}_2\text{O}$ has 2 bonding pairs and 2 lone pairs; intense lone pair - lone pair and lone pair - bonding pair repulsions compress the H-O-H bond angle further to $104.5^\circ$ . | [1]   |
| 68 | C   | Option C is correct. In gaseous $\text{BeCl}_2$ , the central beryllium atom has 2 valence electrons, forming two single Be-Cl bonding pairs and having 0 lone pairs. Two electron pairs arrange themselves at maximum separation directly opposite each other, producing a linear geometry with a $180^\circ$ bond angle.                                                                                                                                                                                                                                         | [1]   |
| 69 | B   | Option B is correct. The central boron atom has 3 valence electrons, forming 3 B-Cl bonding pairs and 0 lone pairs. Three electron pairs repel to the corners of an equilateral triangle, yielding a trigonal planar shape with $120^\circ$ bond angles.                                                                                                                                                                                                                                                                                                           | [1]   |
| 70 | A   | Option A is correct. Carbon has 4 valence electrons + 2 extra electrons from the $2^-$ charge = 6 electrons available. These form 3 sigma bonds to three oxygen atoms and a delocalised pi system over the three bonds (3 bonding regions, 0 lone pairs on carbon). By VSEPR, three charge clouds repel to maximum separation, producing a trigonal planar geometry with exactly $120^\circ$ O-C-O bond angles.                                                                                                                                                    | [1]   |

| Q  | KEY | EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MARKS |
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| 71 | D   | Option D is correct. Sulfur has 6 valence electrons: it forms 2 double bonds (or one double, one dative) with two oxygen atoms (2 bonding regions) and retains 1 non-bonding lone pair. Three electron domains adopt a trigonal planar electron geometry. The non-bonding lone pair exerts greater repulsion on the bonding pairs, squeezing the O-S-O bond angle slightly below 120° to approximately 119°, giving a bent (non-linear) molecular shape.                                                                         | [1]   |
| 72 | C   | Option C is correct. The central phosphorus atom has 5 valence electrons, forming 5 P-Cl bonding pairs and 0 lone pairs (expanding its octet to 10 electrons). Five electron pairs arrange themselves in a trigonal bipyramidal geometry: three equatorial chlorine atoms arranged in a trigonal plane with 120° bond angles, and two axial chlorine atoms at 90° to the equatorial plane.                                                                                                                                       | [1]   |
| 73 | A   | Option A is correct. Sulfur has 6 valence electrons, all shared with six fluorine atoms to form 6 S-F bonding pairs and 0 lone pairs (expanded octet with 12 valence electrons). Six electron pairs arrange themselves towards the vertices of a regular octahedron, giving an octahedral geometry where all adjacent F-S-F bond angles are exactly 90°.                                                                                                                                                                         | [1]   |
| 74 | C   | Option C is correct. In SO <sub>4</sub> <sup>2-</sup> , the central sulfur atom is bonded to four oxygen atoms via four equivalent bonding regions with 0 non-bonding lone pairs on sulfur. Four electron charge clouds repel equally to the vertices of a regular tetrahedron, giving a tetrahedral geometry with bond angles of 109.5°.                                                                                                                                                                                        | [1]   |
| 75 | B   | Option B is correct. With 5 electron pairs, the electron geometry is trigonal bipyramidal. To minimize severe 90° lone pair - bonding pair repulsions, both lone pairs occupy equatorial positions (where they have 120° angles to other equatorial pairs). The remaining three Cl-F bonds occupy the two axial positions and one equatorial position, producing a T-shaped molecular geometry with F-Cl-F bond angles of ~87.5°.                                                                                                | [1]   |
| 76 | D   | Option D is correct. With 6 electron pairs, the electron geometry is octahedral. To minimize the powerful lone pair - lone pair repulsion, the two non-bonding lone pairs position themselves 180° directly opposite each other (axial positions). The four Xe-F bonding pairs lie in a single plane at 90° to each other, yielding a square planar molecular geometry.                                                                                                                                                          | [1]   |
| 77 | C   | Option C is correct. The methyl carbon (C1) is surrounded by 4 single bonding pairs (three C-H and one C-O) and 0 lone pairs, giving a regular tetrahedral geometry with bond angles of ~109.5°. The oxygen atom is surrounded by 2 bonding pairs (one C-O and one O-H) and 2 non-bonding lone pairs, giving a bent (non-linear) geometry with a bond angle of ~104.5° due to lone pair repulsion.                                                                                                                               | [1]   |
| 78 | A   | Option A is correct. Nitrogen is smaller and more electronegative (3.04) than phosphorus (2.19). In NH <sub>3</sub> , the bonding electron pairs are drawn close to the nitrogen nucleus, crowding together and repelling each other strongly to maintain a 107° angle. In PH <sub>3</sub> , the bonding pairs are located further away from the larger phosphorus nucleus, reducing mutual bonding-pair repulsion and allowing the lone pair to compress the bond angle down to 93.5° (approaching pure 90° p-orbital bonding). | [1]   |
| 79 | D   | Option D is correct. Nitrogen has 5 valence electrons minus 1 electron for the positive charge = 4 electrons. It forms two double bonds with the two oxygen atoms (O=N+=O) with 0 lone pairs on nitrogen. With two double-bond charge clouds repelling to maximum distance, NO <sub>2</sub> <sup>+</sup> is strictly linear with a 180° bond angle (isoelectronic with CO <sub>2</sub> ).                                                                                                                                        | [1]   |
| 80 | B   | Option B is correct. The central iodine atom has 7 valence electrons + 1 extra electron from the negative charge + 2 shared from two outer I atoms = 10 valence electrons (5 electron pairs: 2 bonding pairs and 3 lone pairs). In a trigonal bipyramidal arrangement, all three lone pairs occupy equatorial positions at 120° to minimize repulsion, leaving the two I-I bonds in the axial positions at 180°, giving a linear molecular geometry.                                                                             | [1]   |
| 81 | B   | Option B is the accurate scientific description: at any given instant, the continuous random motion of electrons within an electron cloud may produce an asymmetrical charge distribution, generating a temporary instantaneous dipole. This temporary dipole exerts an electrostatic force on the electrons of a neighbouring molecule, inducing an aligned dipole and resulting in weak, short-range attraction.                                                                                                               | [1]   |
| 82 | D   | Option D is correct. Descending Group 18, the number of electrons per atom increases (He: 2 to Xe: 54). Larger electron clouds with outer electrons situated further from the nucleus are held less tightly and are more readily distorted (polarisable). This creates larger instantaneous and induced dipoles, leading to stronger London dispersion forces that require more thermal energy to overcome.                                                                                                                      | [1]   |
| 83 | A   | Option A is correct. Both isomers have identical numbers of electrons (42). Straight-chain pentane has an extended rod-like shape that provides a large surface area of contact, allowing neighbouring molecules to pack closely and maximize London dispersion forces. 2,2-dimethylpropane has a spherical, branched shape with a smaller surface area of contact, preventing close packing and resulting in weaker dispersion forces and a much lower boiling point.                                                           | [1]   |
| 84 | C   | Option C is correct. Propanone has a strongly polar carbonyl group (C=O, delta+ on C, delta- on O) and an asymmetric structure, possessing a permanent molecular dipole. Because it has no O-H, N-H, or F-H bonds, it cannot form hydrogen bonds with itself; its primary intermolecular attractions are permanent dipole-dipole forces combined with London forces. Methane and helium are non-polar (only London forces). Water's predominant attraction is hydrogen bonding.                                                  | [1]   |
| 85 | A   | Option A is correct. A hydrogen bond requires: (1) a hydrogen atom attached to a small, highly electronegative atom with strong polarising power (specifically N, O, or F), which strips electron density from H leaving it as a bare, highly concentrated positive pole (delta+), and (2) a lone pair of electrons on an adjacent N, O, or F atom with which the exposed H(delta+) interacts electrostatically.                                                                                                                 | [1]   |
| 86 | D   | Option D is correct. Oxygen is much smaller and more electronegative (3.44) than sulfur (2.58). As a result, water molecules form strong intermolecular hydrogen bonds (~20 kJ/mol). Sulfur is not electronegative enough to support hydrogen bonding; H <sub>2</sub> S molecules are held together only by much weaker permanent dipole-dipole and London dispersion forces. Boiling water requires breaking these strong hydrogen bonds, requiring vastly more thermal energy.                                                 | [1]   |

| Q   | KEY | EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | MARKS |
|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 87  | B   | Option B is correct. In H <sub>2</sub> O, each molecule has 2 hydrogen atoms and 2 lone pairs on oxygen (a 1:1 stoichiometric ratio), enabling a continuous 3D network where each H <sub>2</sub> O molecule participates in an average of 4 hydrogen bonds. In HF, each molecule has only 1 hydrogen atom despite having 3 lone pairs on fluorine (hydrogen-deficient); thus, it can form an average of only 2 hydrogen bonds per molecule. The double number of hydrogen bonds in water outweighs the slightly greater strength of individual H-F bonds. | [1]   |
| 88  | C   | Option C is correct. In solid ice, each water molecule is hydrogen-bonded to 4 neighbouring water molecules in a rigid tetrahedral arrangement. This creates an open hexagonal crystalline framework with relatively large empty spaces/cavities. When ice melts at 0 °C, the thermal kinetic energy breaks some of these hydrogen bonds (~15%), allowing water molecules to tumble into the vacant cavities and pack closer together, making liquid water denser than ice.                                                                               | [1]   |
| 89  | B   | Option B is correct. Ethanol contains an O-H group with an active hydrogen atom attached to oxygen, enabling molecules to form strong intermolecular hydrogen bonds. In methoxymethane (an ether), both methyl groups are bonded to the central oxygen (C-O-C); there are no hydrogen atoms directly bonded to oxygen, so it cannot form hydrogen bonds with itself. It is held only by weaker dipole-dipole and London forces, boiling at -24.8 °C.                                                                                                      | [1]   |
| 90  | A   | Option A is correct. The hydroxyl group (-OH) of an alcohol can act as both a hydrogen-bond donor (via H delta+) and a hydrogen-bond acceptor (via lone pairs on O) to form favorable intermolecular hydrogen bonds with surrounding water molecules. For small alcohols (methanol, ethanol, propanol), these exothermic interactions readily overcome the hydrogen bonds between water molecules and London forces between alcohol molecules.                                                                                                            | [1]   |
| 91  | D   | Option D is correct. As the non-polar hydrocarbon chain lengthens, its hydrophobic character dominates. Inserting a bulky C <sub>6</sub> H <sub>13</sub> chain into water requires breaking many strong hydrogen bonds between water molecules. The London dispersion forces formed between the alkyl chain and water are too weak to compensate for this energy cost, making dissolution thermodynamically unfavorable.                                                                                                                                  | [1]   |
| 92  | C   | Option C is correct. The enthalpy of vaporization of water (40.7 kJ/mol) represents the energy required to overcome intermolecular hydrogen bonds separating molecules. The atomisation of water into atoms requires breaking two covalent O-H bonds (2 x 464 = 928 kJ/mol). This demonstrates that intermolecular forces are weak physical interactions (~5% of the strength of a true covalent bond).                                                                                                                                                   | [1]   |
| 93  | A   | Option A is correct. Halogens exist as non-polar diatomic molecules (X <sub>2</sub> ). Descending Group 17, the number of electrons increases: Cl <sub>2</sub> (34 e-) < Br <sub>2</sub> (70 e-) < I <sub>2</sub> (106 e-). The larger, more diffuse electron clouds are more readily polarised, generating stronger instantaneous-induced dipole attractions (London dispersion forces). This increases melting and boiling points, transitioning from gas (Cl <sub>2</sub> ) to liquid (Br <sub>2</sub> ) to solid (I <sub>2</sub> ).                   | [1]   |
| 94  | C   | Option C is correct. Complementary base pairing in DNA is maintained by intermolecular hydrogen bonds between specific functional groups (N-H...O and N-H...N). Because a G-C base pair is linked by three hydrogen bonds compared to only two in an A-T pair, DNA strands with high GC content require more thermal energy to break the hydrogen bonds, resulting in a higher melting temperature (T <sub>m</sub> ).                                                                                                                                     | [1]   |
| 95  | B   | Option B is correct. Viscosity measures a fluid's resistance to flow, which depends on the magnitude of internal intermolecular friction. Propan-1-ol has only one -OH group. Glycerol has three -OH groups per molecule, allowing each molecule to form multiple hydrogen bonds in all directions with neighbouring molecules. This creates an extensive, highly cohesive 3D hydrogen-bonding network that strongly resists molecular shear, making glycerol very viscous.                                                                               | [1]   |
| 96  | D   | Option D is correct. Boron is in Group 13 and has 3 valence electrons. It forms three single covalent bonds with three fluorine atoms, each contributing 1 electron. Therefore, boron is surrounded by exactly 3 x 2 = 6 valence electrons. It has an incomplete octet (electron-deficient) and possesses a vacant 2p orbital, allowing it to act as a powerful Lewis acid.                                                                                                                                                                               | [1]   |
| 97  | C   | Option C is correct. Nitrogen is in Period 2 (valence shell n=2), possessing only 2s and 2p orbitals (maximum 4 orbitals, accommodating at most 8 electrons; octet rule cannot be exceeded). Phosphorus is in Period 3 (valence shell n=3) and has vacant, low-lying 3d orbitals in addition to 3s and 3p. It can promote a 3s electron into a 3d orbital to form five sp <sup>3</sup> d hybrid orbitals, accommodating 10 bonding electrons (expanded octet).                                                                                            | [1]   |
| 98  | A   | Option A is correct. In CO <sub>2</sub> (O=C=O), carbon forms two double bonds with oxygen. Each double bond contains 2 electron pairs = 4 electrons, giving 8 shared bonding electrons in total. Each oxygen atom retains two non-bonding lone pairs (2 x 2 = 4 lone pairs = 8 non-bonding electrons). Carbon has 0 lone pairs. All atoms satisfy the octet rule.                                                                                                                                                                                        | [1]   |
| 99  | D   | Option D is correct. Nitrogen has 5 valence electrons + 1 electron from the negative charge = 6 valence electrons. It forms one N=O double bond (sharing 2 electrons) and one N-O single bond (sharing 1 electron, or one dative bond). Total electrons used in bonding = 4. The remaining 2 electrons form exactly 1 non-bonding lone pair on the central nitrogen atom. This lone pair gives the nitrite ion its bent shape (~115°).                                                                                                                    | [1]   |
| 100 | B   | Option B is false (making it the correct answer to the question). Once a coordinate (dative) bond forms, the shared electron pair is completely indistinguishable from any other covalent bond; all bonds around the central atom have identical bond lengths, strengths, and properties (as seen in NH <sub>4</sub> <sup>+</sup> and H <sub>3</sub> O <sup>+</sup> ). Statements B, C, and D are all fundamentally true.                                                                                                                                 | [1]   |
| 101 | C   | Option C is correct. In Al <sub>2</sub> Cl <sub>6</sub> , two monomeric AlCl <sub>3</sub> molecules join together. Each aluminium atom has an empty 3p orbital (6 valence electrons). Two chlorine atoms (one from each AlCl <sub>3</sub> unit) each donate a non-bonding lone pair into the empty 3p orbital of the opposite aluminium atom, creating two dative covalent bridging bonds (Al-Cl->Al). Thus, there are exactly 2 coordinate bonds, with chlorine acting as the donor.                                                                     | [1]   |
| 102 | A   | Option A is correct. Although one bond is formally formed by dative donation of the nitrogen lone pair into the empty 1s orbital of H <sup>+</sup> , once the bond is established, electrons are indistinguishable and completely delocalised across the four equivalent sp <sup>3</sup> hybrid orbitals. All four N-H bonds have identical bond lengths (103 pm) and bond enthalpies (391 kJ/mol), forming a regular tetrahedron (109.5°).                                                                                                               | [1]   |

| Q   | KEY | EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | MARKS |
|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 103 | D   | Option D is correct. All three molecules possess 4 valence electron pairs around the central atom. The repulsive strength follows the order: lone pair – lone pair > lone pair – bonding pair > bonding pair – bonding pair. CH <sub>4</sub> has 0 lone pairs (109.5°); NH <sub>3</sub> has 1 lone pair, which compresses the H–N–H angle by ~2.5° to 107°; H <sub>2</sub> O has 2 lone pairs, whose mutual repulsion compresses the H–O–H angle by another ~2.5° to 104.5°.                                                                                                           | [1]   |
| 104 | B   | Option B is correct. In SF <sub>6</sub> , sulfur forms 6 single covalent bonds with six fluorine atoms, accommodating 6 x 2 = 12 valence electrons in its outer shell (expanded octet using 3s, 3p, and 3d orbitals). CCl <sub>4</sub> and NF <sub>3</sub> have complete octets of 8 electrons. BF <sub>3</sub> has an incomplete octet of 6 electrons (electron-deficient).                                                                                                                                                                                                           | [1]   |
| 105 | A   | Option A is correct. In ice, every water molecule forms 4 hydrogen bonds to four neighbouring molecules in a regular tetrahedral arrangement, creating an open, cage-like hexagonal crystal lattice with significant void spaces. Upon melting at 0 °C, some hydrogen bonds break, allowing water molecules to tumble closer together into the voids, increasing the liquid density.                                                                                                                                                                                                   | [1]   |
| 106 | C   | Option C is correct. In water (H <sub>2</sub> O), there are 2 hydrogen atoms and 2 lone pairs on oxygen, perfectly matched in a 1:1 ratio so that every molecule can form an average of 4 hydrogen bonds, building a robust 3D network. In HF, there is only 1 hydrogen atom despite 3 lone pairs on fluorine, so hydrogen bonding is stoichiometrically limited to an average of only 2 hydrogen bonds per molecule (linear chains), requiring much less thermal energy to vaporize.                                                                                                  | [1]   |
| 107 | B   | Option B is correct. Let us count all bonds in propyne (CH <sub>3</sub> –C≡C–H): Three C–H single bonds on the methyl group = 3 sigma; One C–C single bond = 1 sigma; One C≡C triple bond = 1 sigma and 2 pi; One terminal C–H single bond = 1 sigma. Total sigma bonds = 3 + 1 + 1 + 1 = 6 sigma bonds. Total pi bonds = 2 pi bonds.                                                                                                                                                                                                                                                  | [1]   |
| 108 | D   | Option D is correct. In CCl <sub>4</sub> , the C–Cl bonds are polar because chlorine is more electronegative than carbon. However, CCl <sub>4</sub> has a regular tetrahedral geometry with 109.5° bond angles. The four identical C–Cl dipole vectors cancel out completely in 3D space, resulting in zero net dipole moment. In H <sub>2</sub> O (bent), NH <sub>3</sub> (trigonal pyramidal), and SO <sub>2</sub> (bent), the presence of lone pairs breaks spatial symmetry, leaving permanent net dipoles.                                                                        | [1]   |
| 109 | A   | Option A is correct. Lattice enthalpy is proportional to $(q_1 \times q_2) / (r_+ + r_-)$ . For MgO, the charges are +2 and -2 (charge product = 4), and both ions are smaller (Mg <sup>2+</sup> : 72 pm, O <sup>2-</sup> : 140 pm). For NaCl, the charges are +1 and -1 (charge product = 1), and both ions are larger (Na <sup>+</sup> : 102 pm, Cl <sup>-</sup> : 181 pm). The fourfold charge factor and shorter inter-ionic distance make the lattice enthalpy of MgO (-3791 kJ/mol) nearly 5 times larger than NaCl (-787 kJ/mol), requiring vastly more thermal energy to melt. | [1]   |
| 110 | C   | Option C is correct. Phase changes (such as boiling water) involve overcoming weak intermolecular forces (hydrogen bonds in water, Delta H <sub>vap</sub> ~ 40.7 kJ/mol), leaving the covalent molecules intact. Breaking the intramolecular covalent O–H bonds within water molecules requires an average of 464 kJ/mol. Thus, covalent bonds are roughly 10 to 20 times stronger than intermolecular forces.                                                                                                                                                                         | [1]   |